



# Geometry of Relaxed Fair Regression: A Unified Framework for Aware and Unaware Settings

Marie Generali Lince, Vincent Divol, Rémi Flamary, Solenne Gaucher, Patrick Loiseau

## ► To cite this version:

Marie Generali Lince, Vincent Divol, Rémi Flamary, Solenne Gaucher, Patrick Loiseau. Geometry of Relaxed Fair Regression: A Unified Framework for Aware and Unaware Settings. 2026. <hal-05625006>

**HAL Id: hal-05625006**

**<https://hal.science/hal-05625006v1>**

Preprint submitted on 18 May 2026

**HAL** is a multi-disciplinary open access archive for the deposit and dissemination of scientific research documents, whether they are published or not. The documents may come from teaching and research institutions in France or abroad, or from public or private research centers.

L'archive ouverte pluridisciplinaire **HAL**, est destinée au dépôt et à la diffusion de documents scientifiques de niveau recherche, publiés ou non, émanant des établissements d'enseignement et de recherche français ou étrangers, des laboratoires publics ou privés.



Distributed under a Creative Commons CC BY 4.0 - Attribution - International License

# Geometry of Relaxed Fair Regression: A Unified Framework for Aware and Unaware Settings

Marie Generali Lince<sup>\*1, 2, 4</sup>, Vincent Divol<sup>†2</sup>, Rémi Flamary<sup>‡3</sup>, Solenne Gaucher<sup>§3</sup>, and Patrick Loiseau<sup>¶4</sup>

<sup>1</sup>Inria, Soda team

<sup>2</sup>CREST, ENSAE, IP Paris

<sup>3</sup>CMAP, CNRS, Ecole Polytechnique, Institut Polytechnique de Paris

<sup>4</sup>Inria, Fairplay joint team

May 17, 2026

## Abstract

Fairness-accuracy trade-offs are a central concern in the deployment of fairness-aware machine learning methods. When sensitive attributes are unavailable at inference time—the so called unawareness setting, principled methods for obtaining accurate predictions under relaxed fairness constraints are largely missing. In this work, we address this gap by formulating regression under a demographic parity penalty as an optimal transport problem. Our framework unifies both the *aware* and *unaware* settings and characterizes optimal prediction functions via optimal transport maps, under both squared Wasserstein-2 and Total Variation penalties. These results reveal that the choice of penalty reflects fundamentally different fairness philosophies: the Wasserstein penalty induces a smooth, population-wide compromise, while Total Variation enforces exact parity for a subset of individuals. Building on these theoretical characterizations, we propose an algorithm that is simple to implement, computationally efficient, and consistently matches or outperforms state-of-the-art baselines on real-world benchmarks.

## 1 Introduction

As machine learning models are increasingly deployed in high-stakes domains, mitigating algorithmic bias has become a critical objective. A significant portion of the fairness literature addresses group fairness, which involves imposing constraints relative to a sensitive attribute. Among these, Demographic Parity (DP) has received considerable attention due to its simplicity and interpretability, aiming to ensure that predictions are statistically independent of the sensitive attribute. Mitigation strategies typically intervene on the training data (*pre-processing*), the learning algorithm (*in-processing*), or the model’s predictions (*post-processing*) [4, 13]. Post-processing methods are particularly attractive as they allow for the retrofitting of any black-box predictor without requiring retraining, making them highly applicable in real-world settings where the base model is already deployed. In this work, we focus on post-processing for regression tasks under the DP constraint.

---

<sup>\*</sup>marie.general@ensae.fr

<sup>†</sup>vincent.divol@ensae.fr

<sup>‡</sup>remi.flamary@polytechnique.edu

<sup>§</sup>solenne.gaucher@polytechnique.edu

<sup>¶</sup>patrick.loiseau@inria.fr

The strategy for enforcing DP depends on data availability at inference, yielding two distinct paradigms: the *aware* setting, where the sensitive attribute is observed alongside standard features, and the *unaware* setting, where it is unavailable at inference time. A classical heuristic to circumvent this constraint is to explicitly estimate the sensitive attribute using a proxy classifier (i.e., a plug-in estimator) and apply group-aware corrections, but this inevitably propagates estimation errors. Furthermore, enforcing *exact* DP in either setting is often too rigid, causing severe degradation in predictive utility. This limitation motivates the study of *relaxed* fairness, which allows for controlled violations to optimize the accuracy-fairness trade-off.

**Related works.** Empirical Risk Minimization (ERM) under fairness penalties or relaxed fairness constraints is common within the *in-processing* paradigm. Numerous methods directly incorporate fairness penalties or bounded constraints into the learning objective during model training [1, 5, 20, 22, 25, 33, 10]. More recently, optimal transport (OT) has been leveraged within this paradigm to regularize latent representations and enforce relaxed fairness even when sensitive attributes are unobserved [23]. These approaches inherit a core limitation of in-processing methods: they presuppose direct access to, and control over, the algorithm’s training procedure. In many real-world settings, however, the algorithm is a black box or otherwise unavailable, and post hoc correction of its predictions becomes necessary. Then, achieving relaxed fairness through *post-processing* requires shifting from gradient-based latent optimization to geometric transformations of the final predictions, avoiding pitfalls of arbitrary relaxations [27].

OT provides a natural geometric framework for enforcing DP through post-processing approaches. In the *aware* setting, exact fairness is achieved by mapping group marginal distributions of the Bayes predictor to their Wasserstein barycenter [9, 7, 19, 18, 11, 5, 25], and closed-form continuous relaxations exist to control the trade-off [8]. In the *unaware* setting, where demographics are unobserved at test time, achieving parity often relies on noisy proxies or robust optimization techniques [6, 3]. Acting directly in the continuous space, recent work has theoretically characterized the optimal regression function under exact DP using mappings over 2D measures [12]. However, extending continuous geometric relaxations to the unaware setting remains an open challenge. State-of-the-art methods for relaxed unaware fairness, such as FairReg [31], currently bypass exact continuous geometry relying on grid-based discretization and stochastic optimization. This highlights a significant gap in the literature: while exact unaware fairness admits an OT formulation, its relaxed counterpart is solved via grid-based approximations.

**Outline and contributions.** In this paper, we bridge this gap by establishing exact pointwise closed-form reductions for relaxed fairness across both *aware* and *unaware* settings, focusing on the two-groups case. While restrictive, the two-groups assumption is common in the literature [26] and captures the practically important scenario in which one group corresponds to a protected class, while already revealing key phenomena. Our main contributions are as follows:

- **A unified framework for relaxed fair regression:** In Section 2.2, we formulate a general, penalized OT objective that unifies relaxed fair regression across both aware and unaware settings, accommodating different fairness relaxation metrics ( $\mathcal{W}_2$  and TV).
- **Closed-form solutions:** In Section 2.3, we provide the first exact, continuous geometric reductions for relaxed unaware fairness under both  $\mathcal{W}_2$  and TV penalties. We provide closed-form expressions of the optimal almost-fair prediction functions based on the solution of an OT problem.
- **Practical implications:** In Section 2.4, we make explicit the trade-offs inherent in the choice of fairness penalty and contrast their practical implications. We show that the relaxation penalty is not just a numerical convenience, but reflects fundamentally different approaches to fairness.

- **Implementation and experiments:** In Section 3, we demonstrate how these closed-form solutions can be efficiently estimated from finite samples to build continuous, out-of-sample fair predictors. In Section 4, our theory-driven approach is empirically validated on the Law School and Communities and Crime datasets. The code is provided in the supplement and will be made public upon publication.

**Notation.** Let  $\mathcal{P}(\Omega)$  be the set of probability measures on  $\Omega$ , and  $\mathcal{L}(\cdot)$  the law of a random variable. For  $\mu \in \mathcal{P}(\mathbb{R})$ , denote its cumulative distribution function (c.d.f.) by  $F_\mu$ . For a measurable map  $T$ ,  $T_\# \mu$  is the pushforward measure. The set of couplings with marginals  $\mu$  and  $\nu$  is denoted  $\Pi(\mu, \nu)$ . The OT cost between  $\mu$  and  $\nu$  under a cost  $c$  is defined by  $\text{OT}_c(\mu, \nu) = \inf_{\pi \in \Pi(\mu, \nu)} \int c(x, y) d\pi(x, y)$ .  $\mathcal{W}_2^2(\mu, \nu)$  is the squared Wasserstein-2 distance, i.e.,  $\text{OT}_{c_2}(\mu, \nu)$  with cost  $c_2(x, y) = (x - y)^2$ , and  $\text{TV}(\mu, \nu)$  the Total Variation distance, i.e.  $\text{OT}_{c_0}(\mu, \nu)$  with cost  $c_0(x, y) = \mathbb{1}_{x \neq y}$  [24, Proposition 4.7]. Finally, for  $a \in \mathbb{R}$ ,  $a_+ = \max(a, 0)$  and  $a_- = -\min(a, 0)$ .

## 2 Unified framework for aware and unaware relaxed fair regression

We consider a regression problem where individuals are characterized by a non-sensitive feature  $X$  taking values in some set  $\mathcal{X} \subset \mathbb{R}^d$ , a binary sensitive attribute  $S \in \{+, -\}$ , and a continuous target variable  $Y \in \mathbb{R}$  with  $\mathbb{E}[Y^2] < \infty$ . The law of  $X$  in group  $S = +$  is denoted  $\chi^+ = \mathcal{L}(X|S = +)$ , and the law of  $X$  in group  $S = -$  is denoted  $\chi^- = \mathcal{L}(X|S = -)$ . Slightly abusing notations, let us denote the conditional expectation of  $Y$  given  $(X, S)$  as  $\eta(x, s) = \mathbb{E}[Y | X = x, S = s]$ , and the conditional expectation of  $Y$  given  $X$  as  $\eta(x) = \mathbb{E}[Y | X = x]$ . Finally, we define  $p^+ = \mathbb{P}(S = +)$  and  $p^- = \mathbb{P}(S = -)$  the probabilities of the two groups.

In regression problems, the goal is to predict a continuous outcome  $Y \in \mathbb{R}$  from either the pair  $(X, S)$  (in the awareness setting) or from  $X$  (in the unawareness setting). Then, under mean square error, the Bayes optimal predictor is given by the conditional expectation  $\eta$ . Unfortunately, its predictions often reflect systemic biases toward  $S$  [4]. Among the criteria characterizing fair prediction, DP is satisfied by a prediction function  $f$  if its predictions are statistically independent of  $S$ :  $(f(X, S) \perp\!\!\!\perp S)$  in the awareness setting, and  $(f(X) \perp\!\!\!\perp S)$  in the unawareness setting. Equivalently, DP holds if  $\mathcal{L}(f(X, S) | S = +) = \mathcal{L}(f(X, S) | S = -)$  (resp.  $\mathcal{L}(f(X) | S = +) = \mathcal{L}(f(X) | S = -)$ ), meaning that the conditional distributions of predictions are identical across demographic groups.

### 2.1 Existing results for exact fairness

**Exact fairness under awareness.** In the *aware* setting, the sensitive attribute  $S$  is observed with  $X$  at prediction time. In this setting, Chzhen et al. [9] show that finding the optimal fair predictor is equivalent to solving the following Wasserstein barycenter problem. For  $s \in \{+, -\}$ , let  $\mu_\eta^s = \mathcal{L}(\eta(X, S) | S = s)$ . When the distributions  $\mu_\eta^+$  and  $\mu_\eta^-$  are non atomic, they show that the distribution of the optimal fair prediction  $\nu^*$  corresponds to the Wasserstein barycenter

$$\nu^* = \arg \min_{\nu \in \mathcal{P}(\mathbb{R})} \{p^+ \mathcal{W}_2^2(\mu_\eta^+, \nu) + p^- \mathcal{W}_2^2(\mu_\eta^-, \nu)\}. \quad (1)$$

Since the distributions  $\mu_\eta^+$  and  $\mu_\eta^-$  are one-dimensional, the problem admits a closed-form OT map in terms of quantiles. The optimal fair prediction  $f^*(x, s)$  is obtained by applying this map (transporting from  $\mu_\eta^s$  to the barycenter  $\nu^*$ ) to  $\eta(x, s)$ . We detail the closed-form solution in Appendix B.

**Exact fairness under unawareness.** In the *unaware* setting,  $S$  is unobserved at inference time. As established by Divol and Gaucher [12], the optimal fair prediction can only infer  $S$  implicitly and bases its predictions on the signed group probability  $\Delta$  given by

$$\Delta(x) = \frac{\mathbb{P}(S = + | X = x)}{p^+} - \frac{\mathbb{P}(S = - | X = x)}{p^-}. \quad (2)$$

Let us denote  $((\chi^+ - \chi^-)_+, (\chi^+ - \chi^-)_-)$  the normalized Jordan decomposition of the signed measure  $\chi^+ - \chi^-$ , where the normalization constant is chosen so that  $(\chi^+ - \chi^-)_+$  and  $(\chi^+ - \chi^-)_-$  are probability measures<sup>1</sup>. Then, the fair regression problem reduces to an OT problem between the probability measures  $\mu_{\eta, \Delta}^+$  and  $\mu_{\eta, \Delta}^-$  on  $\mathbb{R} \times \mathbb{R}$ , where

$$\mu_{\eta, \Delta}^+ = (\eta(\cdot), \Delta(\cdot))_{\#} (\chi^+ - \chi^-)_+ \quad \text{and} \quad \mu_{\eta, \Delta}^- = (\eta(\cdot), \Delta(\cdot))_{\#} (\chi^+ - \chi^-)_-.$$

Note that, by construction, the probability measure  $\mu_{\eta, \Delta}^+$  (resp.  $\mu_{\eta, \Delta}^-$ ) only gives weight to the half space  $\{(h, d) \in \mathbb{R}^2 : d > 0\}$  (resp. the half space  $\{(h, d) \in \mathbb{R}^2 : d < 0\}$ ): indeed, for any measurable set  $\mathcal{E}$  such that  $\Delta(x) < 0$  for all  $x \in \mathcal{E}$ , it must be that  $(\chi^+ - \chi^-)_+(\mathcal{E}) = 0$ . Reducing the fair regression problem to an OT problem requires the following assumption.

**Assumption 1.** The measures  $\mu_{\eta, \Delta}^+$  and  $\mu_{\eta, \Delta}^-$  give zero mass to graphs of functions in the sense that for any measurable function  $F : \mathbb{R}^* \rightarrow \mathbb{R}$  and  $s \in \{+, -\}$ ,  $\mu_{\eta, \Delta}^s(\{(F(d), d) : d \neq 0\}) = 0$ .

Under Assumption 1, Divol and Gaucher [12] shows that finding the optimal fair predictor reduces to solving the following barycenter problem

$$\inf_{\nu \in \mathcal{P}(\mathbb{R})} \left\{ \text{OT}_{c_u}(\mu_{\eta, \Delta}^+, \nu) + \text{OT}_{c_u}(\mu_{\eta, \Delta}^-, \nu) \right\}, \quad (P_{\infty})$$

with cost  $c_u(z, y) = \frac{|h-y|^2}{|d|}$  specific to the unaware geometry, where  $z = (h, d) \in \mathbb{R} \times \mathbb{R}$ . Note that this is a two-to-one dimensional OT problem, which in general does not admit a closed-form solution but can be solved using numerical methods [30].

**Reducing awareness to unawareness.** As noted by Divol and Gaucher [12], the results for the awareness setting can be recovered from those in the unawareness setting by assuming that the sensitive attribute  $S$  is included in the features  $X$ , or equivalently that  $S$  is  $X$ -measurable. In this case,  $\Delta(x) = \frac{1}{p^+}$  if  $S = +$ , and  $\Delta(x) = -\frac{1}{p^-}$  if  $S = -$ . Then, Assumption 1 reduces to assuming non-atomicity of the measures  $\mu_{\eta}^s$ , a condition that is necessary for the results of Chzhen et al. [9] to hold. Similarly, the transport cost simplifies as  $c((\eta(x), \Delta(x)), y) = p^+(\eta(x) - y)^2$  if  $S = +$ , and  $c((\eta(x), \Delta(x)), y) = p^-(\eta(x) - y)^2$  if  $S = -$ . Consequently, the OT problem defined in the unawareness setting (Eq.  $(P_{\infty})$ ) reduces, in the awareness setting, to the problem given in Eq. (1). For this reason, we focus in the remainder of the paper on the more general unawareness setting, noting that the awareness case follows as a direct special case of our results.

*Remark 2.1* (Noisy attributes). Our formulation also covers the case where a proxy  $\hat{S}$  is available at inference time. Indeed, including  $\hat{S}$  as a covariate in the feature vector  $X$ , we obtain  $\Delta(x, \hat{s}) = \frac{\mathbb{P}(S=+|X=x, \hat{S}=\hat{s})}{p^+} - \frac{\mathbb{P}(S=-|X=x, \hat{S}=\hat{s})}{p^-}$ , thus interpolating between the two settings. When  $\hat{S}$  is either  $X$ -measurable or contains uninformative noise,  $\mathbb{P}(S|X, \hat{S}) = \mathbb{P}(S|X)$ , recovering the unaware formulation. Conversely, when  $\hat{S} = S$ , we recover the aware setting [8].

<sup>1</sup>See Divol and Gaucher [12], Section 3.2.1 for a construction of these measures.

## 2.2 Optimal Transport for relaxed fairness: a unified framework

In many practical settings, enforcing strict DP can significantly degrade predictive accuracy, sometimes to an unacceptable extent. In this section, we present our general framework for relaxed fair regression. As established in Section 2.1, DP is satisfied if and only if the pushforwards of the positive and negative variations of the signed measure  $\chi^+ - \chi^-$  are equal. To achieve relaxed fairness, we penalize the risk of a prediction function by its  $\mathcal{D}$ -unfairness defined below.

**Definition 1** ( $\mathcal{D}$ -unfairness). Let  $\mathcal{D}$  be a probability discrepancy measure over  $\mathcal{P}(\mathbb{R})$ . The  $\mathcal{D}$ -unfairness of a prediction function  $f$  is defined as

$$\mathcal{U}_{\mathcal{D}}(f) = \mathcal{D}(f_{\#}(\chi^+ - \chi^-)_+, f_{\#}(\chi^+ - \chi^-)_-).$$

The choice of unfairness measure  $\mathcal{U}_{\mathcal{D}}$  enables us to cast the relaxed regression problem as an optimal transport problem. In particular, it upper bounds the  $\mathcal{D}$  distance between the distributions of the predictions:  $\mathcal{U}_{\mathcal{D}}(f) \geq \mathcal{D}(\mathcal{L}(f(X)|S=+), \mathcal{L}(f(X)|S=-))$ . Although this bound is generally not tight, under awareness those terms become equal (see Appendix A.5).

We consider two choices for the discrepancy measure  $\mathcal{D}$ : the squared Wasserstein-2 distance ( $\mathcal{U}_{\mathcal{W}_2^2}$ ), which evaluates the quadratic displacement required to align the distributions, and the Total Variation distance ( $\mathcal{U}_{TV}$ ), which evaluates the maximal probability mass of individuals receiving disparate predictions. Both penalties can be expressed as  $\mathcal{D}(\nu^+, \nu^-) = \text{OT}_{c_{\mathcal{D}}}(\nu^+, \nu^-)$  for some cost function  $c_{\mathcal{D}}$ , equal to  $c_2(y, y') = (y - y')^2$  for  $\mathcal{W}_2^2$  and to  $c_0(y, y') = \mathbb{1}_{\{y \neq y'\}}$  for TV. Our objective is to find the almost-fair prediction function  $f_{\lambda, \mathcal{D}}^*$  which solves the following penalized risk minimization problem:

$$f_{\lambda, \mathcal{D}}^* \in \arg \min_f \mathbb{E} \left[ (f(X) - \eta(X))^2 \right] + \lambda m \mathcal{U}_{\mathcal{D}}(f) \quad (3)$$

where  $m = \text{TV}(\chi^+, \chi^-)/2$  is a normalization constant. Driving  $\lambda \rightarrow \infty$  enforces  $\mathcal{U}_{\mathcal{D}}(f) = 0$ , which is equivalent to enforcing exact DP.

**Relaxed fair regression as an OT problem.** To solve (3), we adopt an OT perspective. Under Assumption 1, we show that when  $\mathcal{D}$  is taken to be either  $\mathcal{W}_2^2$  or TV, the value of (3) coincides with that of the following OT problem:

$$\inf_{\nu^+, \nu^- \in \mathcal{P}(\mathbb{R})} \left\{ \text{OT}_{c_u}(\mu_{\eta, \Delta}^+, \nu^+) + \text{OT}_{c_u}(\mu_{\eta, \Delta}^-, \nu^-) + \lambda \text{OT}_{c_{\mathcal{D}}}(\nu^+, \nu^-) \right\} \quad (P_{\lambda, \mathcal{D}})$$

where the cost  $c_{\mathcal{D}}$  is such that  $\mathcal{D}(\nu^+, \nu^-) = \text{OT}_{c_{\mathcal{D}}}(\nu^+, \nu^-)$ .

**Theorem 1.** Let  $\mathcal{D}(\cdot, \cdot) = \text{OT}_{c_{\mathcal{D}}}(\cdot, \cdot)$  for  $c_{\mathcal{D}}(y, y') = (y - y')^2$  or  $c_{\mathcal{D}}(y, y') = \mathbb{1}_{\{y \neq y'\}}$ . Under Assumption 1, Problem  $(P_{\lambda, \mathcal{D}})$  is solved by deterministic transport maps  $T_{\lambda, \mathcal{D}}^+$  and  $T_{\lambda, \mathcal{D}}^-$  which push  $\mu_{\eta, \Delta}^+$  and  $\mu_{\eta, \Delta}^-$  to the minimizing measures  $\nu^+$  and  $\nu^-$ , respectively. Moreover, the solution of the relaxed fair regression problem (3) is given by

$$f_{\lambda, \mathcal{D}}^*(x) = \begin{cases} T_{\lambda, \mathcal{D}}^+(\eta(x), \Delta(x)) & \text{if } \Delta(x) > 0, \\ T_{\lambda, \mathcal{D}}^-(\eta(x), \Delta(x)) & \text{if } \Delta(x) < 0, \\ \eta(x) & \text{else.} \end{cases} \quad (4)$$

Theorem 1 shows that the relaxed fair regression problem reduces to solving Problem  $(P_{\lambda, \mathcal{D}})$  and finding an OT map. The corresponding problems arising in the unawareness and awareness settings are illustrated in Fig. 1. In the awareness setting, the solution amounts to constructing transport maps between one-dimensional distributions. In the unawareness setting, by contrast, one must transport two-dimensional measures  $\mu_{\eta, \Delta}^{\pm}$  to a one dimensional prediction, where each

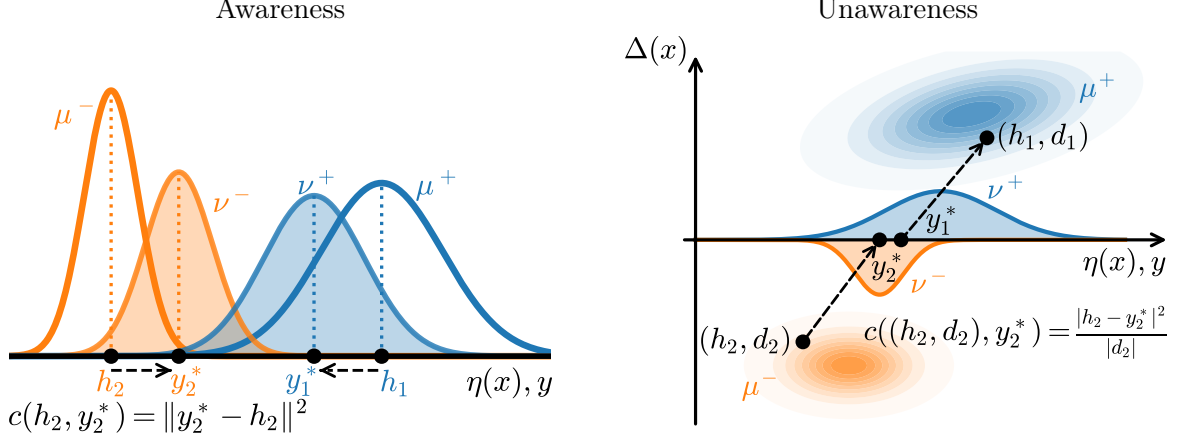


Figure 1: **Geometry of the OT maps.** *Left (Aware):* 1D-to-1D transport mapping group-conditional distributions to almost-fair predictions. *Right (Unaware):* 2D-to-1D transport mapping the joint distributions of  $(\eta(X), \Delta(X))$  under  $\mu_{\eta, \Delta}^+$  and  $\mu_{\eta, \Delta}^-$  to almost-fair predictions.

|                                              | Unaware Setting                                                                                                                                               | Aware Setting                                                                                                                                                |
|----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Objective</b>                             | $\text{OT}_{c_u}(\mu_{\eta, \Delta}^+, \nu^+) + \text{OT}_{c_u}(\mu_{\eta, \Delta}^-, \nu^-) + \lambda \text{OT}_{c_{\mathcal{D}}}(\nu^+, \nu^-)$             | $p^+ \text{OT}_{c_2}(\mu_{\eta}^+, \nu^+) + p^- \text{OT}_{c_2}(\mu_{\eta}^-, \nu^-) + \lambda \text{OT}_{c_{\mathcal{D}}}(\nu^+, \nu^-)$                    |
| <b><math>\mathcal{W}_2^2</math>-fairness</b> | <b>OT-U <math>\mathcal{W}_2</math></b> (Proposition 1)<br><b>Cost:</b> $c_u(z, y) = (h - y)^2 /  d $<br><b>Penalty:</b> $c_{\mathcal{D}}(y, y') = (y - y')^2$ | <b>OT-A <math>\mathcal{W}_2</math></b> ([8], Corollary 1)<br><b>Cost:</b> $c_2(y, y') = (y - y')^2$<br><b>Penalty:</b> $c_{\mathcal{D}}(y, y') = (y - y')^2$ |
| <b>TV-fairness</b>                           | <b>OT-U TV</b> (Proposition 2)<br><b>Cost:</b> $c_u(z, y) = (h - y)^2 /  d $<br><b>Penalty:</b> $c_{\mathcal{D}}(y, y') = \mathbb{1}_{\{y \neq y'\}}$         | <b>OT-A TV</b> (Corollary 2)<br><b>Cost:</b> $c_2(y, y') = (y - y')^2$<br><b>Penalty:</b> $c_{\mathcal{D}}(y, y') = \mathbb{1}_{\{y \neq y'\}}$              |

Table 1: Summary of the unified OT framework. The general problem  $P_{\lambda, \mathcal{D}}$  instantiates into specific pointwise closed-form reductions depending on error cost  $c \in \{c_u, c_2\}$  and geometric penalty  $c_{\mathcal{D}}$ .

point  $(h, d)$  in the support of  $\mu_{\eta, \Delta}^{\pm}$  encodes the Bayes prediction  $\eta$  and the signed group probability  $\Delta$ .

In Table 1, we detail how the general framework in  $(P_{\lambda, \mathcal{D}})$  instantiates for both the aware and unaware settings and fairness measure considered. Interestingly, it leads to three novel relaxed fair regression formulations, and recovers one existing formulation **OT-A  $\mathcal{W}_2$**  that was already proposed in [8].

### 2.3 New results for relaxed fairness

Theorem 1 establishes that the solution to the fair relaxation problem is given by the OT maps solving  $(P_{\lambda, \mathcal{D}})$ . In this section, we show that  $(P_{\lambda, \mathcal{D}})$  can itself be reduced to an OT problem between  $\mu_{\eta, \Delta}^+$  and  $\mu_{\eta, \Delta}^-$  with an appropriately chosen cost. Specializing to the  $\mathcal{W}_2^2$  and the TV-fairness penalty, we obtain solutions to (3) expressed explicitly in terms of the OT maps. Closed-form solutions in the awareness setting follow as a corollary.

**Relaxation under  $\mathcal{W}_2^2$ -fairness.** We consider the fairness penalty  $\mathcal{W}_2^2(\cdot, \cdot) = \text{OT}_{c_2}(\cdot, \cdot)$  with  $c_2(y, y') = (y - y')^2$ . The following proposition reduces  $(P_{\lambda, \mathcal{W}_2})$  to an OT problem between  $\mu_{\eta, \Delta}^+$  and  $\mu_{\eta, \Delta}^-$  with a suitable cost, and gives an expression for the optimal almost-fair prediction via the OT plan. The proof is in Appendix A.3.

**Proposition 1 (OT-U  $\mathcal{W}_2$ ).** *Under Assumption 1, Problem  $(P_{\lambda, \mathcal{W}_2})$  is equivalent to the OT problem*

$$\pi^* \in \arg \min_{\pi \in \Pi(\mu_{\eta, \Delta}^+, \mu_{\eta, \Delta}^-)} \int_{\mathbb{R}^2 \times \mathbb{R}^2} C_{\lambda, c_2}^u(z_1, z_2) d\pi(z_1, z_2), \quad (5)$$

where  $C_{\lambda, c_2}^u(z_1, z_2) = \frac{\lambda}{1 + \lambda(|d_1| + |d_2|)} (h_1 - h_2)^2$  for  $z_1 = (h_1, d_1)$  and  $z_2 = (h_2, d_2)$ . Moreover, a solution to the almost-fair regression problem in (3) is given by Eq. (4) with

$$T_{\lambda, \mathcal{W}_2^2}^+(z_1) = \frac{(1 + \lambda|d_2|)h_1 + \lambda|d_1|h_2}{1 + \lambda(|d_1| + |d_2|)} \quad \text{and} \quad T_{\lambda, \mathcal{W}_2^2}^-(z_2) = \frac{\lambda|d_2|h_1 + (1 + \lambda|d_1|)h_2}{1 + \lambda(|d_1| + |d_2|)} \quad (6)$$

for all pairs  $(z_1, z_2)$  in the support of  $\pi^*$ .

Proposition 1 provides a roadmap for building almost-fair classifiers: first, solve the OT problem (5), then compute the prediction function via the OT plan using Eq. (4) and Eq. (6). Note that while (5) is solved by a (non-deterministic) transport plan, we show in Appendix A.3 that the value defining  $T_{\lambda, \mathcal{W}_2^2}^+(z_1)$  does not depend on the choice of  $z_2$  with  $(z_1, z_2)$  in the support of  $\pi^*$ , so that  $T_{\lambda, \mathcal{W}_2^2}^+$  is a well-defined map (and similarly for  $T_{\lambda, \mathcal{W}_2^2}^-$ ). As noted in Appendix B and Remark A.1, this formulation encompasses prior results as boundary cases. Specifically, in the strict fairness limit ( $\lambda \rightarrow \infty$ ), Proposition 1 recovers, in the awareness framework, the fair regression function of Chzhen et al. [9], and in the unawareness framework, the OT map of Divol and Gaucher [12]. For generic  $\lambda > 0$ , the awareness framework recovers the geodesic interpolations of Chzhen and Schreuder [8], while the unawareness framework yields a novel characterization of the optimal predictor under a relaxed fairness constraint.

**Relaxation under TV-fairness** We now consider the fairness penalty  $\text{TV}(\cdot, \cdot) = \text{OT}_{c_0}(\cdot, \cdot)$  for  $c_0(y, y') = \mathbb{1}_{\{y \neq y'\}}$ . The following proposition reduces  $(P_{\lambda, \text{TV}})$  to an OT problem between  $\mu_{\eta, \Delta}^+$  and  $\mu_{\eta, \Delta}^-$  with a suitable cost, and gives an expression for the optimal almost-fair prediction via the OT plan. The proof is in Appendix A.4.

**Proposition 2 (OT-U TV).** *Under Assumption 1, Problem  $(P_{\lambda, \text{TV}})$  is equivalent to the OT problem*

$$\pi^* \in \arg \min_{\pi \in \Pi(\mu_{\eta, \Delta}^+, \mu_{\eta, \Delta}^-)} \int_{\mathbb{R}^2 \times \mathbb{R}^2} C_{\lambda, c_0}^u(z_1, z_2) d\pi(z_1, z_2), \quad (7)$$

where  $C_{\lambda, c_0}^u(z_1, z_2) = \min\left(\lambda, \frac{(h_1 - h_2)^2}{|d_1| + |d_2|}\right)$ . Moreover, a solution to the fair regression problem in Eq. (3) is given by Eq. (4) with

$$\begin{cases} T_{\lambda, \text{TV}}^+(z_1) = T_{\lambda, \text{TV}}^-(z_2) = \frac{|d_2|h_1 + |d_1|h_2}{|d_1| + |d_2|} & \text{if } \frac{(h_1 - h_2)^2}{|d_1| + |d_2|} \leq \lambda, \\ T_{\lambda, \text{TV}}^+(z_1) = h_1, \quad T_{\lambda, \text{TV}}^-(z_2) = h_2 & \text{if } \frac{(h_1 - h_2)^2}{|d_1| + |d_2|} > \lambda, \end{cases} \quad (8)$$

for all pairs  $z_1 = (h_1, d_1)$  and  $z_2 = (h_2, d_2)$  in the support of  $\pi^*$ .

Here again, while Problem (7) is solved by a (non-deterministic) transport plan, we show in Appendix A.4 that  $T_{\lambda, \text{TV}}^+(z_1)$  is well-defined in the sense that its value does not depend on the choice of  $z_2$  with  $(z_1, z_2)$  in the support of  $\pi^*$  (and similarly for  $T_{\lambda, \text{TV}}^-(z_2)$ ). In the strict fairness limit ( $\lambda \rightarrow \infty$ ), Proposition 1 also recovers previous results [9, 12]. For  $\lambda > 0$ , it yields a novel characterization of the optimal predictor under a TV-fairness penalty in the unawareness framework. In the next section, we discuss the implications of this relaxation and contrast it with the  $\mathcal{W}_2^2$  fairness penalty.



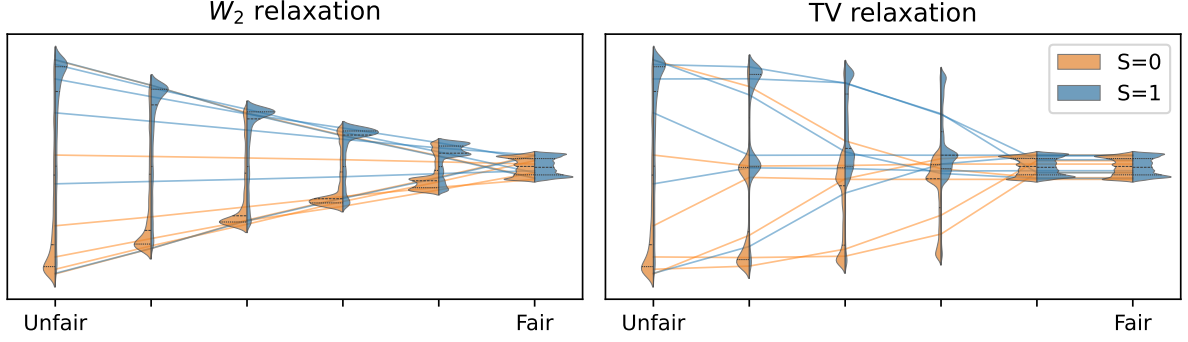


Figure 2: **Individual prediction trajectories for  $\mathcal{W}_2$  vs. TV Relaxation.** Evolution of marginal distributions and subset of individual prediction paths from Unfair (Bayes) to strictly Fair (100%-0% of initial unfairness). The lines show  $\mathcal{W}_2$  induces proportional shift for all individuals, while TV leaves predictions unchanged (horizontal lines) until a hard threshold triggers abrupt merging.

## 2.4 The implicit philosophies of relaxed fairness

The characterizations of the optimal almost-fair prediction function in Propositions 1 and 2 highlight how the choice of fairness penalty impacts the behaviour of fair algorithms. In this section, we contrast the two philosophies induced by the geometry of the  $\mathcal{W}_2$  and TV penalties.

**Homogeneous gap reduction under Wasserstein relaxation.** The  $\mathcal{W}_2$  penalty penalizes the magnitude of the discrepancy in the label space. Therefore, the optimal strategy consists in displacing continuously all predictions in the two groups toward the  $\mathcal{W}_2$  barycenter. This implies that it is preferable for all individuals to be slightly moved together toward the fair prediction than for a few to experience massive disparity. Philosophically, this aligns with an *egalitarian* view of fairness [28], which treats inequality as a shared societal burden to be mitigated globally across the population.

**Heterogeneous exact parity under Total Variation relaxation.** The TV penalty penalizes non-overlapping probability mass, ignoring distances in the prediction space. Compromising by moving predictions slightly closer yields no benefit unless they are merged perfectly. As shown in Proposition 2, this results in a hard-thresholding behavior. This implies that TV enforces exact parity where it is “affordable” (i.e., where the cost is low), and accepts total unfairness for the extremes of the distribution with low densities. TV ensures that a maximized subset of the population achieves perfect indistinguishability. Ethically, this reflects a *sufficientarian* approach [17], focusing on securing a plateau of absolute equality for the majority rather than penalizing outliers.

**Illustration of the paradigms.** We illustrate the effect of the  $\mathcal{W}_2$  and TV relaxations on a synthetic dataset for the unaware setting (Fig. 2) using the implementation detailed in Section 3. Observing individual predictions’ paths across relaxation levels reveals the distinct mechanisms of each penalty. As the fairness constraint tightens, the  $\mathcal{W}_2$  penalty shifts the distributions uniformly, with individual predictions smoothly and linearly interpolating toward the center. By contrast, the TV case exhibits  $L_1$  hard-thresholding. Predictions either remain at their unconstrained values (horizontal paths) or abruptly merge. Furthermore, a saturation effect occurs: for sufficiently large  $\lambda$ , the solution stabilizes to the strict fairness mapping. Full data generation details, explicit visualizations of the mapping geometries, and examples in the aware setting with similar behaviors are provided in Appendix E.1.

### 3 Algorithmic implementation

In the previous section, we expressed the solution of the almost-fair regression problem in terms of a theoretical OT problem. Building on these characterizations, we now propose a practical algorithm to estimate the optimal almost-fair prediction function. We assume access to an estimator  $\hat{\eta}$  of the Bayes regression function  $\eta$ , and an estimator  $\hat{\mathbb{P}}(S = s|X = x)$  of the posterior group probabilities  $\mathbb{P}(S = s|X = x)$ , which can be obtained using standard regression and probabilistic classification methods, and to an unlabeled dataset  $(x_i, s_i)_{i \leq n}$ . The pseudocode is provided in Appendix C.

**Estimation of  $\hat{\Delta}(x)$ .** We first build estimates  $\hat{p}^+$ ,  $\hat{p}^-$  of  $p^+$  and  $p^-$  using  $(x_i, s_i)_{i \leq n}$ , and define  $\hat{\Delta}(x) = \frac{\hat{\mathbb{P}}(S=+|X=x)}{\hat{p}^+} - \frac{\hat{\mathbb{P}}(S=-|X=x)}{\hat{p}^-}$ . We then apply a threshold  $\tau$  to partition training indices into positive ( $\mathcal{I}^+ = \{i : \hat{\Delta}(x_i) > \tau\}$ ) and negative ( $\mathcal{I}^- = \{i : \hat{\Delta}(x_i) < -\tau\}$ ) subsets based on the sign of  $\hat{\Delta}$ , forming the discrete supports of our empirical measures. Crucially, to account for finite-sample estimation errors, we normalize the local weights ( $\hat{d}_i \propto \hat{\Delta}(x_i)$  for  $i \in \mathcal{I}^+$  and  $\hat{d}_i \propto -\hat{\Delta}(x_i)$  for  $i \in \mathcal{I}^-$ ) to enforce mass conservation ( $\sum_{i \in \mathcal{I}^+} |\hat{d}_i| = \sum_{i \in \mathcal{I}^-} |\hat{d}_i| = 1$ ), ensuring the downstream transport problem remains well-posed.

**Discrete OT and barycentric projection.** From the empirical measures  $(\hat{\eta}_i, \hat{d}_i)_{i \in \mathcal{I}^+}$  and  $(\hat{\eta}_j, \hat{d}_j)_{j \in \mathcal{I}^-}$ , we construct the pairwise cost matrix  $\mathbf{C}_\lambda \in \mathbb{R}_+^{|\mathcal{I}^+| \times |\mathcal{I}^-|}$  using the closed-form solutions derived in Proposition 1 (for the  $\mathcal{W}_2$  relaxation) or Proposition 2 (for the TV relaxation).

We then solve the discrete OT problem to find the optimal coupling matrix  $\mathbf{\Pi}$ . For any point  $i \in \mathcal{I}^s$ , its fair pseudo-label  $\hat{f}_i$  is obtained via the barycentric projection of the target mapping. With a slight abuse of notation, let  $T_{\lambda, \mathcal{D}}^s(z_i, z_j)$  denote the closed-form target mapping evaluated for the coupled pair  $(z_i, z_j)$  (derived via Eq. (6) for  $\mathcal{W}_2$  and Eq. (8) for TV). The pseudo-label is then computed as

$$\hat{f}_i = \frac{1}{|\hat{d}_i|} \sum_{j \in \mathcal{I}^{(-s)}} \Pi_{i,j} T_{\lambda, \mathcal{D}}^s((\hat{\eta}_i, \hat{d}_i), (\hat{\eta}_j, \hat{d}_j)).$$

**Continuous mapping and final predictor.** For out-of-sample inference time, we concatenate training points  $(x_i)_{i \leq n}$  and fit a regressor  $\hat{f}_{\text{fair}}$  to predict  $(\hat{f}_i)_{i \leq n}$ . Predicting the almost-fair outcome for a new instance  $x$  only requires evaluating the values  $\hat{\eta}(x)$ ,  $\hat{\Delta}(x)$ , and  $\hat{f}_{\text{fair}}(\hat{\eta}(x), \hat{\Delta}(x))$ .

**Computational complexity.** Our framework’s complexity is dominated by the  $O(n^3 \log n)$  discrete OT step, computed using the network simplex implementation from the POT library [16]. Despite this worst-case bound, bypassing the stochastic gradient descent of Taturyan et al. [31] makes our deterministic approach efficient. Empirically, it runs in seconds for  $n \sim 10^4$ , significantly outperforming FairReg (Appendix G), and can be further accelerated via mini-batching [14].

## 4 Numerical experiments

In this section, we empirically evaluate the performance of our closed-form OT relaxations on both synthetic and real-world datasets.

### 4.1 Baseline and state-of-the-art comparison

We benchmark our methods against the unconstrained (and unfair) Empirical Risk Minimizer (ERM) and the state-of-the-art method for unaware relaxed fair regression, FairReg [31].

**The FairReg algorithm.** To address relaxed unaware fairness, Taturyan et al. [31] propose a post-processing framework that learns a randomized prediction rule  $\pi$  over a finite grid  $\mathcal{T}$ . Instead of a deterministic mapping, they solve an entropy-regularized problem:  $\min_{\pi} \{\mathbb{E}[(\hat{Y}_{\pi} - \hat{\eta}(X))^2] + \frac{1}{\beta} \mathbb{E}[\Psi(\pi)]\}$  subject to  $\mathcal{U}_{\text{KS}}^T(\hat{Y}_{\pi}, S) \leq \epsilon$ , where  $\hat{Y}_{\pi} \sim \pi(\cdot|X)$ ,  $\Psi$  is the negative entropy, and  $\mathcal{U}_{\text{KS}}^T$  is a discretized Kolmogorov-Smirnov (KS) distance. This non-differentiable problem is solved via a dual formulation and stochastic gradient method. Details on the algorithm and its implementation are in Appendix D.1. We compare our unaware OT algorithm against FairReg using their exact experimental protocol.

**Aware and proxy-based baselines.** To establish the empirical lower bound for the accuracy-fairness trade-off, we include the aware optimal transport models (OT-A  $\mathcal{W}_2$  and OT-A TV), which have access to the true sensitive attribute  $S$  at both training and inference time. We also benchmark against a naive plug-in estimator (OT  $\hat{S}$   $\mathcal{W}_2$ ). A common heuristic [3, 6, 21] for the unaware setting is to predict a proxy  $\hat{S}$  using a probabilistic classifier and apply the classical 1D aware OT mapping. Comparing these approaches highlights the cost of classification errors compared to our 2D framework.

**Evaluation metrics.** We evaluate predictive utility via Mean Squared Error (MSE). To quantify fairness, we report three metrics. First, the Wasserstein-2 ( $\mathcal{W}_2$ ) and Total Variation (TV) unfairness. Then, to ensure fair comparison with the FairReg baseline, we report  $\text{KS}^T$ , which approximates the KS distance by computing the maximum absolute c.d.f. difference across a discrete 50-bin grid  $\mathcal{T}$ .

## 4.2 Real-world datasets

To evaluate our framework against the FairReg baseline [31], we use real-world datasets from their evaluation protocol: Communities and Crime, and Law School Admissions. Identical experiments and visualizations for a synthetic and the Adult (Census Income) dataset are provided in Appendices E.2 and F respectively, which show similar gains in performance and robustness of our OT algorithms.

**Datasets descriptions.** The Law School Admissions dataset comprises 20,649 student records from 163 U.S. law schools, featuring academic indicators (e.g., LSAT, GPA) and demographic variables. The regression task is to predict the student’s standardized first-year GPA (`zfygpa`). Because the data exhibits racial imbalance (approximately 84.6% White), we follow standard practice and define the binary sensitive attribute  $S$  as White ( $S = 1$ ) versus Non-White ( $S = 0$ ). The Communities and Crime dataset comprises 1,994 U.S. communities characterized by 128 socio-economic and demographic features. The regression task is to predict the violent crime rate per 100,000 population (`ViolentCrimesPerPop`). To construct  $S$ , we identify the majority racial demographic within each community and binarize it into White ( $S = 1$ ) versus non-White ( $S = 0$ ).

**Results.** As shown in Figs. 3 and 4, our unaware OT framework (OT-U  $\mathcal{W}_2$  and OT-U TV) matches or outperforms the baselines. While the naive plug-in baseline (OT  $\hat{S}$   $\mathcal{W}_2$ ) attempts to leverage continuous OT, its reliance on a noisy proxy attribute significantly degrades the accuracy-fairness trade-off, often pushing its curve above FairReg. Conversely, OT-U  $\mathcal{W}_2$  achieves superior trade-offs on the continuous metric and remains highly competitive on KS and TV. Interestingly,  $\mathcal{W}_2$ -penalized models achieve better tradeoffs than TV-based ones *even when evaluated on the TV metric*.

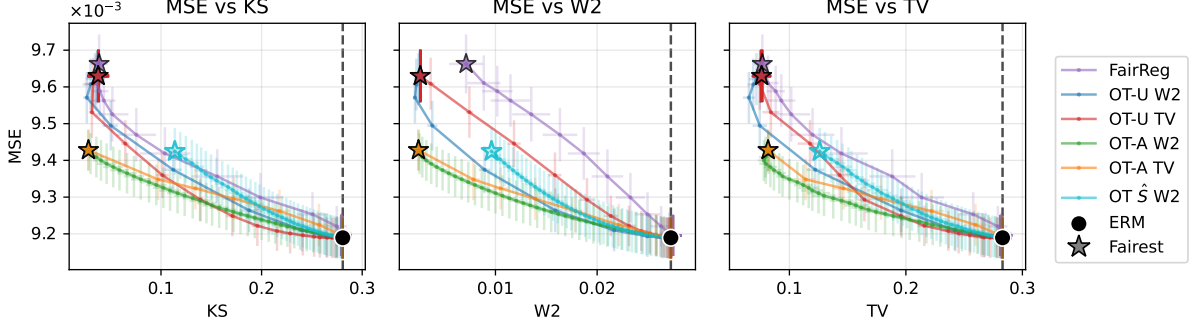


Figure 3: **Accuracy-fairness trade-offs on the Law School dataset.** Relaxation trajectories from the unconstrained ERM (black dot) to the exact fair models (stars). Error bars indicate the standard deviation across 10 random data splits. MSE of the constant predictor is  $10.1 \times 10^{-3}$ .

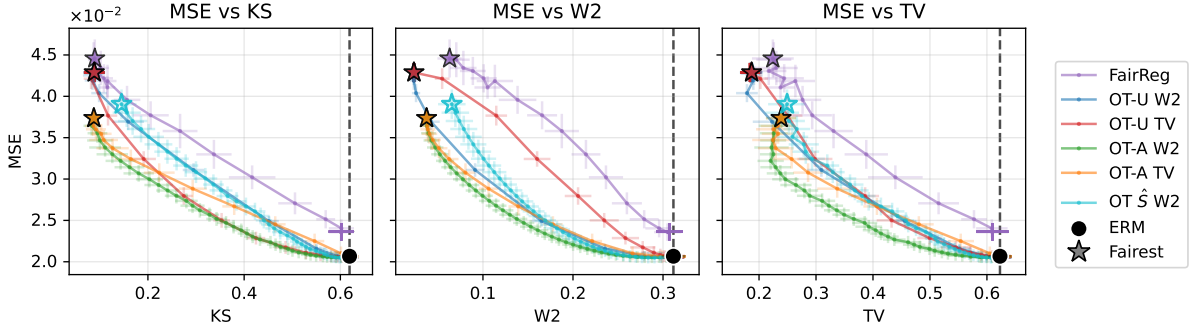


Figure 4: **Accuracy-fairness trade-offs on the Communities dataset.** Relaxation trajectories from the unconstrained ERM to the exact fair models. MSE of the constant predictor is  $5.5 \times 10^{-2}$ .

## 5 Limitations and perspectives

This work introduced a unified geometric framework for relaxed fair regression, where optimal predictors admit closed-form expressions via optimal transport maps. These characterizations are valuable for two reasons: they clarify the implications of the chosen fairness penalties, and they enable a simple, efficient post-processing algorithm for both aware and unaware settings that consistently matches or outperforms state-of-the-art methods.

We identify the following limitations of our framework. First, our results are currently restricted to TV and  $\mathcal{W}_2^2$  fairness penalties. Extending them to penalties  $\text{OT}_c$  with other costs  $c$  could bring fruitful insight to relaxed fairness and inspire new algorithms. Second, our results are restricted to binary sensitive attributes. Extending them to multi-group fairness would require generalizing the Jordan decomposition argument to multi-marginal OT, an interesting but challenging problem. Finally, as a post-processing method, the performance of our algorithm in terms of accuracy and fairness is intrinsically dependent on the estimators  $\eta$  and  $\Delta$ . Deriving statistical guarantees relating the performance of our algorithm to that of these estimators would help ensure that fairness and accuracy are not substantially degraded.

## Acknowledgments and Disclosure of Funding

This work was supported by Hi! PARIS and the ANR/France 2030 program (ANR-23-IACL-0005), as well as by the Fondation de l’Ecole Polytechnique. Additional support was provided by the French National Research Agency (ANR) through grant ANR-23-CE23-0002 and the PEPR IA FOUNDRY project (ANR-23-PEIA-0003).

## References

- [1] Alekh Agarwal, Miroslav Dudik, and Zhiwei Steven Wu. Fair regression: Quantitative definitions and reduction-based algorithms. In Kamalika Chaudhuri and Ruslan Salakhutdinov, editors, *Proceedings of the 36th International Conference on Machine Learning*, volume 97 of *Proceedings of Machine Learning Research*, pages 120–129. PMLR, 09–15 Jun 2019. URL <https://proceedings.mlr.press/v97/agarwal19d.html>.
- [2] Charalambos D. Aliprantis and Kim C. Border. *Infinite Dimensional Analysis: A Hitchhiker’s Guide*. Springer Berlin Heidelberg, 3 edition, 2006. ISBN 978-3-540-32696-0. doi: 10.1007/3-540-29587-9.
- [3] Pranjal Awasthi, Matthäus Kleindessner, and Jamie Morgenstern. Equalized odds post-processing under imperfect group information. In Silvia Chiappa and Roberto Calandra, editors, *Proceedings of the Twenty Third International Conference on Artificial Intelligence and Statistics*, volume 108 of *Proceedings of Machine Learning Research*, pages 1770–1780. PMLR, 26–28 Aug 2020. URL <https://proceedings.mlr.press/v108/awasthi20a.html>.
- [4] Solon Barocas, Moritz Hardt, and Arvind Narayanan. *Fairness and Machine Learning: Limitations and Opportunities*. MIT Press, 2023.
- [5] Richard A. Berk, Hoda Heidari, Shahin Jabbari, Matthew Joseph, Michael Kearns, Jamie Morgenstern, Seth Neel, and Aaron Roth. A convex framework for fair regression. In *Proceedings of the 4th Workshop on Fairness, Accountability, and Transparency in Machine Learning (FAT/ML)*, ICML Workshop, August 2017. URL <https://arxiv.org/abs/1706.02409>. Co-located with KDD 2017. Preprint: arXiv:1706.02409.
- [6] Jiahao Chen, Nathan Kallus, Xiaojie Mao, Geoffrey Svacha, and Madeleine Udell. Fairness under unawareness: Assessing disparity when protected class is unobserved. In *Proceedings of the Conference on Fairness, Accountability, and Transparency*, FAT\* ’19, page 339–348. ACM, January 2019. doi: 10.1145/3287560.3287594. URL <http://dx.doi.org/10.1145/3287560.3287594>.
- [7] Silvia Chiappa, Ray Jiang, Tom Stepleton, Aldo Pacchiano, Heinrich Jiang, and John Aslanides. A general approach to fairness with optimal transport. *Proceedings of the AAAI Conference on Artificial Intelligence*, 34(04):3633–3640, Apr. 2020. doi: 10.1609/aaai.v34i04.5771. URL <https://ojs.aaai.org/index.php/AAAI/article/view/5771>.
- [8] Evgenii Chzhen and Nicolas Schreuder. A minimax framework for quantifying risk-fairness trade-off in regression. *The Annals of Statistics*, 50(4):2416–2442, August 2022. doi: 10.1214/22-AOS2198. URL <https://arxiv.org/abs/2007.14265>.
- [9] Evgenii Chzhen, Christophe Denis, Mohamed Hebiri, Luca Oneto, and Massimiliano Pontil. Fair regression with wasserstein barycenters. In H. Larochelle, M. Ranzato, R. Hadsell, M.F. Balcan, and H. Lin, editors, *Advances in Neural Information Processing Systems*, volume 33, pages 7321–7331. Curran Associates, Inc., 2020. URL [https://proceedings.neurips.cc/paper\\_files/paper/2020/file/51cbbd2611e844ece5d80878eb770436-Paper.pdf](https://proceedings.neurips.cc/paper_files/paper/2020/file/51cbbd2611e844ece5d80878eb770436-Paper.pdf).
- [10] Andrew Cotter, Heinrich Jiang, Maya Gupta, Serena Wang, Taman Narayan, Seungil You, and Karthik Sridharan. Optimization with non-differentiable constraints with applications to fairness, recall, churn, and other goals. *Journal of Machine Learning Research*, 20(172): 1–59, 2019. URL <http://jmlr.org/papers/v20/18-616.html>.
- [11] Christophe Denis, Romuald Elie, Mohamed Hebiri, and François Hu. Fairness guarantees in multi-class classification with demographic parity. *Journal of Machine Learning Research*, 25(130):1–46, 2024. URL <https://jmlr.org/papers/v25/23-0322.html>.

- [12] Vincent Divol and Solenne Gaucher. Demographic parity in regression and classification within the unawareness framework, 2024. URL <https://arxiv.org/abs/2409.02471>.
- [13] Cynthia Dwork, Moritz Hardt, Toniann Pitassi, Omer Reingold, and Richard Zemel. Fairness through awareness. In *Proceedings of the 3rd Innovations in Theoretical Computer Science Conference*, ITCS '12, pages 214–226. ACM, January 2012. doi: 10.1145/2090236.2090255. URL <https://arxiv.org/abs/1104.3913>.
- [14] Kilian Fatras, Younes Zine, Rémi Flamary, Rémi Gribonval, and Nicolas Courty. Learning with minibatch Wasserstein : asymptotic and gradient properties. In *AISTATS 2020 - 23rd International Conference on Artificial Intelligence and Statistics*, volume volume 108 of *PMLR*, pages 1–20, Palermo, Italy, June 2020. URL <https://hal.science/hal-02502329>.
- [15] Eugene A Feinberg, Pavlo O Kasyanov, and Nina V Zadoianchuk. Fatou’s lemma for weakly converging probabilities. *Theory of Probability & Its Applications*, 58(4):683–689, 2014.
- [16] Rémi Flamary, Nicolas Courty, Alexandre Gramfort, Mokhtar Z Alaya, Aurélie Boissunon, Stanislas Chambon, Laetitia Chapel, Adrien Corenflos, Kilian Fatras, Nemo Fournier, et al. Pot: Python optimal transport. *Journal of Machine Learning Research*, 22(78):1–8, 2021.
- [17] Harry Gordon Frankfurt. Equality as a moral ideal. *Ethics*, 98(1):21–43, October 1987. doi: 10.1086/292913. URL <https://www.jstor.org/stable/2381290>.
- [18] Solenne Gaucher, Nicolas Schreuder, and Evgenii Chzhen. Fair learning with wasserstein barycenters for non-decomposable performance measures. In Francisco Ruiz, Jennifer Dy, and Jan-Willem van de Meent, editors, *Proceedings of The 26th International Conference on Artificial Intelligence and Statistics*, volume 206 of *Proceedings of Machine Learning Research*, pages 2436–2459. PMLR, 25–27 Apr 2023. URL <https://proceedings.mlr.press/v206/gaucher23a.html>.
- [19] Thibaut Le Gouic, Jean-Michel Loubes, and Philippe Rigollet. Projection to fairness in statistical learning, 2020. URL <https://arxiv.org/abs/2005.11720>.
- [20] Tatsunori Hashimoto, Megha Srivastava, Hongseok Namkoong, and Percy Liang. Fairness without demographics in repeated loss minimization. In Jennifer Dy and Andreas Krause, editors, *Proceedings of the 35th International Conference on Machine Learning*, volume 80 of *Proceedings of Machine Learning Research*, pages 1929–1938. PMLR, 10–15 Jul 2018. URL <https://proceedings.mlr.press/v80/hashimoto18a.html>.
- [21] Nathan Kallus, Xiaojie Mao, and Angela Zhou. Assessing algorithmic fairness with unobserved protected class using data combination. *Manage. Sci.*, 68(3):1959–1981, March 2022. ISSN 0025-1909. doi: 10.1287/mnsc.2020.3850. URL <https://doi.org/10.1287/mnsc.2020.3850>.
- [22] Preethi Lahoti, Alex Beutel, Jilin Chen, Kang Lee, Flavien Prost, Nithum Thain, Xuezhi Wang, and Ed Chi. Fairness without demographics through adversarially reweighted learning. In H. Larochelle, M. Ranzato, R. Hadsell, M.F. Balcan, and H. Lin, editors, *Advances in Neural Information Processing Systems*, volume 33, pages 728–740. Curran Associates, Inc., 2020. URL [https://proceedings.neurips.cc/paper\\_files/paper/2020/file/07fc15c9d169ee48573edd749d25945d-Paper.pdf](https://proceedings.neurips.cc/paper_files/paper/2020/file/07fc15c9d169ee48573edd749d25945d-Paper.pdf).
- [23] Thibaud Leteno, Michael Perrot, Charlotte Laclau, Antoine Gourru, and Christophe Gravier. Fair text classification via transferable representations. *Journal of Machine Learning Research*, 26(239):1–47, 2025. URL <http://jmlr.org/papers/v26/25-0485.html>.

- [24] D.A. Levin, Y. Peres, and E.L. Wilmer. *Markov Chains and Mixing Times*. American Mathematical Soc., 2009. ISBN 9780821886274. URL <https://books.google.fr/books?id=6Cg5Nq5sSv4C>.
- [25] Zhu Li, Adrian Perez-Suay, Gustau Camps-Valls, and Dino Sejdinovic. Kernel dependence regularizers and gaussian processes with applications to algorithmic fairness, 2019. URL <https://arxiv.org/abs/1911.04322>.
- [26] Zachary Lipton, Julian McAuley, and Alexandra Chouldechova. Does mitigating ml's impact disparity require treatment disparity? In S. Bengio, H. Wallach, H. Larochelle, K. Grauman, N. Cesa-Bianchi, and R. Garnett, editors, *Advances in Neural Information Processing Systems*, volume 31. Curran Associates, Inc., 2018. URL [https://proceedings.neurips.cc/paper\\_files/paper/2018/file/8e0384779e58ce2af40eb365b318cc32-Paper.pdf](https://proceedings.neurips.cc/paper_files/paper/2018/file/8e0384779e58ce2af40eb365b318cc32-Paper.pdf).
- [27] Michael Lohaus, Michaël Perrot, and Ulrike Von Luxburg. Too relaxed to be fair. In *Proceedings of the 37th International Conference on Machine Learning, ICML'20*. JMLR.org, 2020.
- [28] John Rawls. *A Theory of Justice*. Harvard University Press, 1971.
- [29] Filippo Santambrogio. *Optimal Transport for Applied Mathematicians: Calculus of Variations, PDEs, and Modeling*, volume 87 of *Progress in Nonlinear Differential Equations and Their Applications*. Birkhäuser Cham, 1 edition, 2015. ISBN 978-3-319-20827-5. doi: 10.1007/978-3-319-20828-2. Published 27 October 2015.
- [30] Eloi Tanguy, Julie Delon, and Nathaël Gozlan. Computing barycentres of measures for generic transport costs. *arXiv preprint arXiv:2501.04016*, 2024.
- [31] Gayane Taturyan, Evgenii Chzhen, and Mohamed Hebiri. Regression under demographic parity constraints via unlabeled post-processing. In A. Globerson, L. Mackey, D. Belgrave, A. Fan, U. Paquet, J. Tomczak, and C. Zhang, editors, *Advances in Neural Information Processing Systems*, volume 37, pages 117917–117953. Curran Associates, Inc., 2024. URL [https://proceedings.neurips.cc/paper\\_files/paper/2024/file/d5c3ecf397fff63419bb5f5f2d8afe33-Paper-Conference.pdf](https://proceedings.neurips.cc/paper_files/paper/2024/file/d5c3ecf397fff63419bb5f5f2d8afe33-Paper-Conference.pdf).
- [32] Cédric Villani. *Optimal Transport: Old and New*, volume 338 of *Grundlehren der mathematischen Wissenschaften*. Springer Science & Business Media, 2008. ISBN 9783540710509.
- [33] Muhammad Bilal Zafar, Isabel Valera, Manuel Gomez-Rodriguez, and Krishna P. Gummadi. Fairness constraints: A flexible approach for fair classification. *Journal of Machine Learning Research*, 20(75):1–42, 2019. URL <http://jmlr.org/papers/v20/18-262.html>.

# Appendix Table of Contents

---

|                                                                                                                          |           |
|--------------------------------------------------------------------------------------------------------------------------|-----------|
| <b>A Proofs of propositions and lemmas</b>                                                                               | <b>2</b>  |
| A.1 Equivalence between (3) and $(P_{\lambda, \mathcal{D}})$ , assuming deterministic optimal transport maps             | 2         |
| A.2 Reduction of $(P_{\lambda, \mathcal{D}})$ to an OT problem between $\mu_{\eta, \Delta}^+$ and $\mu_{\eta, \Delta}^-$ | 4         |
| A.3 Proof of Proposition 1 (OT-U $\mathcal{W}_2$ )                                                                       | 5         |
| A.4 Proof of Proposition 2 (OT-U TV)                                                                                     | 8         |
| A.5 Properties of the fairness penalty                                                                                   | 10        |
| <b>B Proofs for the relaxed aware setting</b>                                                                            | <b>10</b> |
| B.1 Aware $\mathcal{W}_2$ relaxation                                                                                     | 11        |
| B.2 Aware TV relaxation                                                                                                  | 12        |
| <b>C Algorithm pseudocode</b>                                                                                            | <b>13</b> |
| <b>D Experimental setup and implementation details</b>                                                                   | <b>14</b> |
| D.1 Details on the FairReg Baseline                                                                                      | 14        |
| D.2 General experimental details                                                                                         | 14        |
| <b>E Experiments on synthetic datasets</b>                                                                               | <b>15</b> |
| E.1 Empirical illustration of the geometric relaxations                                                                  | 15        |
| E.2 Impact of confounding strenght and trade-offs                                                                        | 16        |
| <b>F Experiments on real-world datasets</b>                                                                              | <b>18</b> |
| F.1 Accuracy-fairness trade-offs                                                                                         | 20        |
| F.2 Evaluation of the fully constrained models                                                                           | 20        |
| F.3 Robustness across different base estimators                                                                          | 21        |
| <b>G Computational efficiency</b>                                                                                        | <b>23</b> |

---



## A Proofs of propositions and lemmas

**Notation for the proofs.** Throughout the appendix, we define the 2D mapped feature space as  $\mathcal{Z} := \mathbb{R} \times \mathbb{R}$ . An element in this space is denoted as  $z = (h, d) \in \mathcal{Z}$ , where  $h \in \mathbb{R}$  represents the initial base prediction  $\eta(x)$ , and  $d \in \mathbb{R}$  represents the local density ratio  $\Delta(x)$ . For readability, we denote its absolute value as  $a = |d|$ . Additionally, for a measure  $\mu \in \mathcal{P}(\mathbb{R})$ , we denote its generalized quantile function as  $Q_\mu(t) = \inf\{y \in \mathbb{R} : F_\mu(y) \geq t\}$  for  $t \in (0, 1)$ .

We begin the proof of Theorem 1 by showing that if Problem  $(P_{\lambda, \mathcal{D}})$  is solved by transport maps, then the solution to Problem (3) is indeed given as in (4). This is done in Lemma 1. The remainder of the proof will then consist in proving that for the choice of fairness penalties considered, and under Assumption 1, Problem  $(P_{\lambda, \mathcal{D}})$  is indeed solved by transport maps. To do this, we show that Problem  $(P_{\lambda, \mathcal{D}})$  reduces to an OT problem between measures  $\mu_{\eta, \Delta}^+$ ,  $\mu_{\eta, \Delta}^-$  (Lemma 2), before showing in Appendix A.3 and Appendix A.4 that for the  $\mathcal{W}_2^2$  and the TV fairness penalties respectively, this problem is indeed solved by deterministic transport maps. Along the way, we prove Proposition 1 in Appendix A.3, and Proposition 2 in Appendix A.4.

### A.1 Equivalence between (3) and $(P_{\lambda, \mathcal{D}})$ , assuming deterministic optimal transport maps

**Lemma 1.** *If Problem  $(P_{\lambda, \mathcal{D}})$  is solved by deterministic transport maps  $T_{\lambda, \mathcal{D}}^+$  and  $T_{\lambda, \mathcal{D}}^-$  which push  $\mu_{\eta, \Delta}^+$  and  $\mu_{\eta, \Delta}^-$  to the minimizing measures  $\nu^+$  and  $\nu^-$ , respectively, then the solution of the relaxed fair regression problem (3) is given by*

$$f_{\lambda, \mathcal{D}}^*(x) = \begin{cases} T_{\lambda, \mathcal{D}}^+(\eta(x), \Delta(x)) & \text{if } \Delta(x) > 0, \\ T_{\lambda, \mathcal{D}}^-(\eta(x), \Delta(x)) & \text{if } \Delta(x) < 0, \\ \eta(x) & \text{else.} \end{cases}$$

*Proof.* Let us start by setting some additional notations. We denote  $\chi = p^+ \chi^+ + p^- \chi^-$  the law of  $X$ . Let  $|\chi^+ - \chi^-|$  be the variation of  $\chi^+ - \chi^-$ , then for  $m = \frac{\text{TV}(\chi^+, \chi^-)}{2}$ , we define the normalized Jordan decomposition of  $\chi^+ - \chi^-$  through

$$\begin{cases} m(\chi^+ - \chi^-)_+ = \frac{1}{2}(|\chi^+ - \chi^-| + \chi^+ - \chi^-), \\ m(\chi^+ - \chi^-)_- = \frac{1}{2}(|\chi^+ - \chi^-| - \chi^+ + \chi^-), \end{cases}$$

that is  $(m(\chi^+ - \chi^-)_+, m(\chi^+ - \chi^-)_-)$  is the Jordan decomposition of  $\chi^+ - \chi^-$ . Then, we have  $\chi^+ - \chi^- = m(\chi^+ - \chi^-)_+ - m(\chi^+ - \chi^-)_-$ , and

$$\begin{aligned} \Delta(x) &= \frac{\mathbb{P}(S = +|X)}{p^+} - \frac{\mathbb{P}(S = -|X)}{p^-} \\ &= \frac{d(\chi^+ - \chi^-)}{d\chi}(x) \\ &= m \frac{d(\chi^+ - \chi^-)_+}{d\chi}(x) - m \frac{d(\chi^+ - \chi^-)_-}{d\chi}(x). \end{aligned}$$

Finally, we also define  $\mathcal{X}_+ = \{x \in \mathcal{X} : \Delta(x) > 0\}$ ,  $\mathcal{X}_- = \{x \in \mathcal{X} : \Delta(x) < 0\}$ , and  $\mathcal{X}_0 = \{x \in \mathcal{X} : \Delta(x) = 0\}$ .  $\mathcal{X}_+$  is the support of  $(\chi^+ - \chi^-)_+$ , and  $\mathcal{X}_-$  is the support of  $(\chi^+ - \chi^-)_-$ . With

these notations, for all prediction function  $f$ ,

$$\begin{aligned}
\mathbb{E}[(\eta(X) - f(X))^2] &= \int (\eta(x) - f(x))^2 d\chi(x) \\
&= \int_{\mathcal{X}_+} (\eta(x) - f(x))^2 \frac{d\chi}{d(\chi^+ - \chi^-)_+}(x) d(\chi^+ - \chi^-)_+(x) \\
&\quad + \int_{\mathcal{X}_-} (\eta(x) - f(x))^2 \frac{d\chi}{d(\chi^+ - \chi^-)_-}(x) d(\chi^+ - \chi^-)_-(x) \\
&\quad + \int_{\mathcal{X}_=} (\eta(x) - f(x))^2 d\chi(x) \\
&= \int_{\mathcal{X}_+} (\eta(x) - f(x))^2 \frac{m}{|\Delta(x)|} d(\chi^+ - \chi^-)_+(x) \\
&\quad + \int_{\mathcal{X}_-} (\eta(x) - f(x))^2 \frac{m}{|\Delta(x)|} d(\chi^+ - \chi^-)_-(x) \\
&\quad + \int_{\mathcal{X}_=} (\eta(x) - f(x))^2 d\chi(x) \\
&= m \int_{\mathcal{X}_+} c_u((\eta(x), \Delta(x)), f(x)) d(\chi^+ - \chi^-)_+(x) \\
&\quad + m \int_{\mathcal{X}_-} c_u((\eta(x), \Delta(x)), f(x)) d(\chi^+ - \chi^-)_-(x) \\
&\quad + \int_{\mathcal{X}_=} (\eta(x) - f(x))^2 d\chi(x).
\end{aligned}$$

Recall that  $\mu_{\eta, \Delta}^+ = (\eta, \Delta)_\#(\chi^+ - \chi^-)_+$ , and define  $\tilde{\nu}^+ = f_\#(\chi^+ - \chi^-)_+$ . Then,  $(\eta(X), \Delta(X), f(X))$  defines a coupling between  $\mu_{\eta, \Delta}^+$  and  $\tilde{\nu}^+$ , so

$$\int_{\mathcal{X}_+} c_u((\eta(x), \Delta(x)), f(x)) d(\chi^+ - \chi^-)_+(x) \geq \text{OT}_{c_u}(\mu_{\eta, \Delta}^+, \tilde{\nu}^+).$$

Defining similarly  $\tilde{\nu}^- = f_\#(\chi^+ - \chi^-)_-$ , we have

$$\int_{\mathcal{X}_-} c_u((\eta(x), \Delta(x)), f(x)) d(\chi^+ - \chi^-)_-(x) \geq \text{OT}_{c_u}(\mu_{\eta, \Delta}^-, \tilde{\nu}^-).$$

Now, by definition,  $\mathcal{U}_{\mathcal{D}}(f) = \text{OT}_{c_{\mathcal{D}}}(\tilde{\nu}^+, \tilde{\nu}^-)$ , so we find

$$\begin{aligned}
&\min_f \mathbb{E}[(\eta(X) - f(X))^2] + \lambda m \mathcal{U}_{\mathcal{D}}(f) \\
&\geq m \inf_{\tilde{\nu}^+, \nu^- \in \mathcal{P}(\mathbb{R})} \left( \text{OT}_{c_u}(\mu_{\eta, \Delta}^+, \tilde{\nu}^+) + \text{OT}_{c_u}(\mu_{\eta, \Delta}^-, \tilde{\nu}^-) + \lambda \text{OT}_{c_{\mathcal{D}}}(\tilde{\nu}^+, \tilde{\nu}^-) \right). \tag{9}
\end{aligned}$$

Now, if the OT problem on the right hand side of (9) is solved by deterministic transport maps  $T_{\lambda, \mathcal{D}}^+$  and  $T_{\lambda, \mathcal{D}}^-$  which push  $\mu_{\eta, \Delta}^+$  and  $\mu_{\eta, \Delta}^-$  to the minimizing measures  $\nu^+$  and  $\nu^-$ , respectively, then for  $f_{\lambda, \mathcal{D}}^*$  given in Equation (4) we have

$$\begin{aligned}
\int_{\mathcal{X}_+} c_u((\eta(x), \Delta(x)), f_{\lambda, \mathcal{D}}^*(x)) d(\chi^+ - \chi^-)_+(x) &= \text{OT}_{c_u}(\mu_{\eta, \Delta}^+, \nu^+), \\
\int_{\mathcal{X}_-} c_u((\eta(x), \Delta(x)), f_{\lambda, \mathcal{D}}^*(x)) d(\chi^+ - \chi^-)_-(x) &= \text{OT}_{c_u}(\mu_{\eta, \Delta}^-, \nu^-), \\
\int_{\mathcal{X}_=} (\eta(x) - f_{\lambda, \mathcal{D}}^*(x))^2 d\chi(x) &= 0
\end{aligned}$$

so

$$\begin{aligned} & \mathbb{E} [(\eta(X) - f(X)^*)^2] + \lambda m \mathcal{U}_{\mathcal{D}}(f^*) \\ &= m \min_{\tilde{\nu}^+, \tilde{\nu}^-} \left( \text{OT}_{c_u}(\mu_{\eta, \Delta}^+, \tilde{\nu}^+) + \text{OT}_{c_u}(\mu_{\eta, \Delta}^-, \tilde{\nu}^-) + \lambda \text{OT}_{c_{\mathcal{D}}}(\tilde{\nu}^+, \tilde{\nu}^-) \right). \end{aligned}$$

This concludes the proof of Lemma 1.  $\square$

## A.2 Reduction of $(P_{\lambda, \mathcal{D}})$ to an OT problem between $\mu_{\eta, \Delta}^+$ and $\mu_{\eta, \Delta}^-$

As an introductory result, we use the Gluing lemma to show that the minimization problem  $(P_{\lambda, \mathcal{D}})$  can be recast as a simple optimal transport problem for some appropriate cost.

**Lemma 2.** *Let  $c_{\mathcal{D}} : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$  be lower semi-continuous and bounded from below. For any pair of points  $z_1 = (h_1, d_1)$  and  $z_2 = (h_2, d_2)$ , we define the pointwise relaxed cost*

$$C_{\lambda, \mathcal{D}}(z_1, z_2) := \inf_{y_1, y_2 \in \mathbb{R}} \left\{ \frac{(h_1 - y_1)^2}{|d_1|} + \frac{(h_2 - y_2)^2}{|d_2|} + \lambda c_{\mathcal{D}}(y_1, y_2) \right\}. \quad (10)$$

*Then, the problem  $(P_{\lambda, \mathcal{D}})$  is equivalent to the optimal transport problem using this pointwise cost, in the sense that*

$$\inf(P_{\lambda, \mathcal{D}}) = \text{OT}_{C_{\lambda, \mathcal{D}}}(\mu_{\eta, \Delta}^+, \mu_{\eta, \Delta}^-) = \min_{\pi \in \Pi(\mu_{\eta, \Delta}^+, \mu_{\eta, \Delta}^-)} \int_{\mathbb{R} \times \mathbb{R}} C_{\lambda, \mathcal{D}}(z_1, z_2) d\pi(z_1, z_2). \quad (11)$$

Moreover, let  $\pi^*$  be a minimizer of the right-hand side, and let  $(y_1^*, y_2^*) : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}^2$  be a measurable map such that  $(y_1^*(z_1, z_2), y_2^*(z_1, z_2))$  realizes the infimum in (10) for all  $(z_1, z_2) \in \mathbb{R}^2 \times \mathbb{R}^2$ . Then,  $(\nu^{+*}, \nu^{-*}) = (y_1^*, y_2^*)_{\#} \pi^*$  is a minimizer of  $(P_{\lambda, \mathcal{D}})$ .

*Proof of Lemma 2.* To prove this equivalence, we first establish that the pointwise optimization constitutes a lower bound for the global problem, and then demonstrate that this bound is exactly achieved.

**Lower bound using the Gluing lemma.** Let  $\nu^+, \nu^- \in \mathcal{P}(\mathbb{R})$ . The cost  $c_u$  is lower semi-continuous and bounded from below and we assume that the fairness penalty  $c_{\mathcal{D}}$  is also lower semi-continuous and bounded from below. Therefore, by Theorem 1.5 from [29], optimal transport plans exist for all three costs  $c_u$ ,  $c_{\mathcal{D}}$  and  $C_{\lambda, \mathcal{D}}$ . Let  $\gamma_1 \in \Pi(\mu_{\eta, \Delta}^+, \nu^+)$  and  $\gamma_2 \in \Pi(\mu_{\eta, \Delta}^-, \nu^-)$  be optimal transport plans for the cost  $c_u$ , and let  $\gamma_3 \in \Pi(\nu^+, \nu^-)$  be the optimal transport plan for the fairness penalty  $c_{\mathcal{D}}$ .

By the Gluing Lemma (e.g., Lemma 5.5 in Santambrogio [29]), we can construct a joint probability measure  $\rho$  over the product space  $\mathbb{R}^2 \times \mathbb{R}^2 \times \mathbb{R} \times \mathbb{R}$ , with coordinates denoted  $(z_1, z_2, y_1, y_2)$ , such that its respective two-dimensional marginals match  $\gamma_1$ ,  $\gamma_2$ , and  $\gamma_3$ .

The global penalized objective  $(P_{\lambda, \mathcal{D}})$  can then be evaluated as a single integral over this joint measure  $\rho$

$$\int [c_u(z_1, y_1) + c_u(z_2, y_2) + \lambda c_{\mathcal{D}}(y_1, y_2)] d\rho(z_1, z_2, y_1, y_2).$$

For every tuple  $(z_1, z_2, y_1, y_2)$  in the support of  $\rho$ , the integrand is bounded from below by its pointwise infimum over all possible target values  $y_1, y_2 \in \mathbb{R}$  for the given pair  $(z_1, z_2)$ . Substituting this pointwise minimum,  $C_{\lambda, \mathcal{D}}(z_1, z_2)$ , into the integral yields a lower bound. Furthermore, letting  $\pi \in \Pi(\mu_{\eta, \Delta}^+, \mu_{\eta, \Delta}^-)$  denote the projection of  $\rho$  onto its first two variables  $(z_1, z_2)$ , we obtain

$$\text{OT}_{c_u}(\mu_{\eta, \Delta}^+, \nu^+) + \text{OT}_{c_u}(\mu_{\eta, \Delta}^-, \nu^-) + \lambda \text{OT}_{c_{\mathcal{D}}}(\nu^+, \nu^-) \geq \int_{\mathbb{R}^2 \times \mathbb{R}^2} C_{\lambda, \mathcal{D}}(z_1, z_2) d\pi(z_1, z_2).$$

Taking the infimum over all valid source couplings  $\pi$  establishes that  $\min_{\pi} \int C_{\lambda, \mathcal{D}} d\pi$  is a global lower bound for  $(P_{\lambda, \mathcal{D}})$ .

**Existence and achieving the bound.** The cost  $C_{\lambda, \mathcal{D}}$  is lower semi-continuous as an infimum of lower semi-continuous costs. It is also clearly lower bounded since  $c_{\mathcal{D}}$  is bounded from below. Therefore, we can apply Theorem 1.5 of Santambrogio [29], which gives us an optimal coupling  $\pi^* \in \Pi(\mu_{\eta, \Delta}^+, \mu_{\eta, \Delta}^-)$ . We denote the objective function

$$\Phi_{z_1, z_2}(y_1, y_2) := \frac{(h_1 - y_1)^2}{|d_1|} + \frac{(h_2 - y_2)^2}{|d_2|} + \lambda c_{\mathcal{D}}(y_1, y_2).$$

This function is lower semi-continuous and coercive, and therefore has a nonempty, compact set of minimizers. Since  $\Phi_{z_1, z_2}(y_1, y_2)$  is measurable with respect to the variable  $(z_1, z_2)$ , the Measurable Maximum Theorem (see Theorem 18.19 in Aliprantis and Border [2]) applies, which implies that there exists a measurable map  $(y_1^*, y_2^*) : \mathbb{R}^2 \times \mathbb{R}^2 \rightarrow \mathbb{R}^2$  such that  $(y_1^*(z_1, z_2), y_2^*(z_1, z_2))$  is a minimizer of  $\Phi_{z_1, z_2}$  for all  $(z_1, z_2) \in \mathbb{R} \times \mathbb{R}$ .

We can therefore push forward  $\pi^*$  via the map  $(z_1, z_2) \mapsto (z_1, z_2, y_1^*, y_2^*)$  and construct a joint measure  $\rho^*$ . The marginals of  $\rho^*$  over its third and fourth coordinates ( $y_1$  and  $y_2$ ) define target distributions  $\nu^{+*}$  and  $\nu^{-*}$ . Because  $\rho^*$  constitutes a valid joint transport plan for these targets, the optimal transport cost is bounded above by the integral over  $\rho^*$ , yielding

$$\text{OT}_{c_u}(\mu_{\eta, \Delta}^+, \nu^{+*}) + \text{OT}_{c_u}(\mu_{\eta, \Delta}^-, \nu^{-*}) + \lambda \text{OT}_{c_{\mathcal{D}}}(\nu^{+*}, \nu^{-*}) \leq \int_{\mathbb{R}^2 \times \mathbb{R}^2} C_{\lambda, \mathcal{D}}(z_1, z_2) d\pi^*(z_1, z_2).$$

Combined with the previously established lower bound, this proves the equality (11), and that  $\nu^{+*}$  and  $\nu^{-*}$  are minimizers of  $(P_{\lambda, \mathcal{D}})$ .

Moreover, the previous argument clearly shows that for *any* optimal transport plan  $\pi^*$  and *any* choice of measurable minimizers  $(y_1^*, y_2^*)$ , the measures  $\nu^{+*}$  and  $\nu^{-*}$  obtained through the previous construction are minimizers of  $(P_{\lambda, \mathcal{D}})$ , concluding the proof of the lemma.  $\square$

### A.3 Proof of Proposition 1 (OT-U $\mathcal{W}_2$ )

**Proposition 1 (OT-U  $\mathcal{W}_2$ ).** *Under Assumption 1, Problem  $(P_{\lambda, \mathcal{W}_2})$  is equivalent to the OT problem*

$$\pi^* \in \arg \min_{\pi \in \Pi(\mu_{\eta, \Delta}^+, \mu_{\eta, \Delta}^-)} \int_{\mathbb{R}^2 \times \mathbb{R}^2} C_{\lambda, c_2}^u(z_1, z_2) d\pi(z_1, z_2), \quad (5)$$

where  $C_{\lambda, c_2}^u(z_1, z_2) = \frac{\lambda}{1 + \lambda(|d_1| + |d_2|)}(h_1 - h_2)^2$  for  $z_1 = (h_1, d_1)$  and  $z_2 = (h_2, d_2)$ . Moreover, a solution to the almost-fair regression problem in (3) is given by Eq. (4) with

$$T_{\lambda, \mathcal{W}_2}^+(z_1) = \frac{(1 + \lambda|d_2|)h_1 + \lambda|d_1|h_2}{1 + \lambda(|d_1| + |d_2|)} \quad \text{and} \quad T_{\lambda, \mathcal{W}_2}^-(z_2) = \frac{\lambda|d_2|h_1 + (1 + \lambda|d_1|)h_2}{1 + \lambda(|d_1| + |d_2|)} \quad (6)$$

for all pairs  $(z_1, z_2)$  in the support of  $\pi^*$ .

*Proof.* By Lemma 2, the global problem reduces to evaluating the pointwise relaxed cost for  $\mathcal{D}(y_1, y_2) = (y_1 - y_2)^2$ . Fix  $z_i = (h_i, d_i)$  with absolute density weights  $a_i = |d_i| > 0$ , and minimize

$$\Phi_{z_1, z_2}(y_1, y_2) = \frac{(h_1 - y_1)^2}{a_1} + \frac{(h_2 - y_2)^2}{a_2} + \lambda(y_1 - y_2)^2.$$

Because  $\Phi := \Phi_{z_1, z_2}$  is strictly convex with respect to  $y_1$  and  $y_2$ , the optimal pointwise targets are characterized by the first-order optimality conditions, which can be expressed as

$$\begin{cases} \frac{2(y_1 - h_1)}{a_1} + 2\lambda(y_1 - y_2) = 0, \\ \frac{2(y_2 - h_2)}{a_2} - 2\lambda(y_1 - y_2) = 0. \end{cases}$$

Solving the system yields the optimal pointwise targets  $y_1^*$  and  $y_2^*$ . Let  $u := y_1^* - y_2^*$ . From the optimality conditions we have

$$\begin{cases} y_1^* - h_1 = -\lambda a_1 u, \\ y_2^* - h_2 = \lambda a_2 u, \end{cases}$$

Subtracting these identities gives  $u = (h_1 - h_2) - \lambda(a_1 + a_2)u$ , which resolves to  $u = \frac{h_1 - h_2}{1 + \lambda(a_1 + a_2)}$ . On the one hand, plugging this value into the first-order conditions yield

$$\begin{cases} y_1^* &= \frac{h_1(1 + \lambda a_2) + \lambda h_2 a_1}{1 + \lambda(a_1 + a_2)}, \\ y_2^* &= \frac{h_2(1 + \lambda a_1) + \lambda h_1 a_2}{1 + \lambda(a_1 + a_2)}. \end{cases} \quad (12)$$

On the other hand, plugging these back into  $\Phi$  at the optimum yields

$$\begin{aligned} C_{\lambda, c_2}(z_1, z_2) &= \Phi(y_1^*, y_2^*) = \frac{(\lambda a_1 u)^2}{a_1} + \frac{(\lambda a_2 u)^2}{a_2} + \lambda u^2 \\ &= \lambda u^2 (1 + \lambda(a_1 + a_2)) \\ &= \lambda \frac{(h_1 - h_2)^2}{(1 + \lambda(a_1 + a_2))^2} (1 + \lambda(a_1 + a_2)) \\ &= \frac{\lambda}{1 + \lambda(a_1 + a_2)} (h_1 - h_2)^2. \end{aligned}$$

According to Lemma 2, a solution  $(\nu^{+*}, \nu^{-*})$  of  $(P_{\lambda, \mathcal{D}})$  is given by the marginal distributions of  $(y_1^*(Z_1, Z_2), y_2^*(Z_1, Z_2))$  where  $(Z_1, Z_2) \sim \pi^*$  and  $\pi^*$  is as in (5). Our next goal is to show that under Assumption 1, the random variable  $y_1^*(Z_1, Z_2)$  does not depend on  $Z_2$ . To do so, we closely follow the arguments of the Step 2 of Theorem 4 in [12]. The only difference between the two settings is that the cost  $C_{\infty, c_2}(z_1, z_2) = (h_1 - h_2)^2 / (a_1 + a_2)$  is replaced by the cost  $C_{\lambda, c_2}(z_1, z_2)$ . These two costs share the essential property that they are quadratic whenever  $a_1$  and  $a_2$  are fixed, which is enough to make the arguments of the proof in Divol and Gaucher [12] work.

Let us give some additional details. The proof is based on Kantorovich duality, see Section 2.1 in [12] for an introduction to this notion. There exists a function  $\psi$ , called a Kantorovich potential, that satisfies the following properties. Let  $\phi : z_2 \mapsto \sup_{z_1 \in \mathcal{Z}} (\psi(z_1) - C_{\lambda, c_2}(z_1, z_2))$  be the  $C_{\lambda, c_2}$ -transform of  $\psi$ . Then,

1. for all  $z_1$ ,  $\psi(z_1) = \inf_{z_2} (\phi(z_2) + C_{\lambda, c_2}(z_1, z_2))$ ;
2.  $\pi^*$  is supported on the  $C_{\lambda, c_2}^u$ -subdifferential of a Kantorovich potential  $\psi$ , that is the set

$$\Gamma = \{(z_1, z_2) \in \mathcal{Z} \times \mathcal{Z} : \psi(z_1) - \phi(z_2) = C_{\lambda, c_2}(z_1, z_2)\}.$$

The arguments used to prove Lemma 3 in Divol and Gaucher [12] can be reproduced to show that under Assumption 1, the Kantorovich potential  $\psi$  is differentiable in the direction  $h_1$  for  $\mu_{\eta, \Delta}^+$ -almost every  $z_1 = (h_1, d_1)$  (this follows from the fact that, when  $(d_1, d_2)$  is fixed, the cost  $C_{\lambda, c_2}(z_1, z_2)$  is proportional to  $(h_1 - h_2)^2$ ).

Let  $z_1$  be a point in the support of  $\mu_{\eta, \Delta}^+$  where  $\partial_{h_1} \psi$  exists, and let  $z_2$  be such that  $(z_1, z_2)$  is in the support of  $\pi^*$  (and therefore in  $\Gamma$ ). The definitions of  $\phi$  and  $\Gamma$  imply that the function  $z \mapsto \psi(z) - C_{\lambda, c_2}(z, z_2)$  attains its maximum at  $z_1$ . Differentiating, we find that

$$\partial_{h_1} \psi(z_1) = \partial_{h_1} C_{\lambda, c_2}^u(z_1, z_2) = \frac{2\lambda(h_1 - h_2)}{1 + \lambda(a_1 + a_2)} = 2\lambda u.$$

Isolating  $u = \frac{1}{2\lambda} \partial_{h_1} \psi(z_1)$  and substituting it into the optimality condition for  $y_1^*$  gives

$$y_1^*(z_1, z_2) = h_1 - \lambda a_1 u = h_1 - \frac{|d_1|}{2} \partial_{h_1} \psi(z_1).$$

In other words, the value  $y_1^*(z_1, z_2)$  does not depend on the choice of  $z_2$  with  $(z_1, z_2)$  in the support of  $\pi^*$  for  $\mu_{\eta, \Delta}^+$ -almost all  $z_1$ , which is what we wanted to show. A symmetric argument holds for  $y_2^*$ , concluding to the existence of deterministic maps  $T_{\lambda, \mathcal{W}_2^2}^+$  and  $T_{\lambda, \mathcal{W}_2^2}^-$  solving  $(P_{\lambda, \mathcal{D}})$ .

Together with Lemma 1 and Lemma 2, this proves Theorem 1 for the  $\mathcal{W}_2^2$  unfairness penalty. Moreover, the proof of Proposition 1 is concluded by noticing that the expression of the optimal transport maps are given by Eq. (12).  $\square$

Let us note that Proposition 1 implies in particular the result of Theorem 1 in the case  $c_{\mathcal{D}} = c_2$ .

*Remark A.1* (Asymptotic compatibility with exact fairness). As a sanity check, let us verify that the solution of the problem  $(P_{\lambda, \mathcal{D}})$  converge to the solution of the fair regression problem  $(P_{\infty})$  as  $\lambda \rightarrow \infty$ .

Consider the relaxed objective  $(P_{\lambda, \mathcal{D}})$  and its reduced form Eq. (11) with the pointwise cost  $C_{\lambda, c_2}^u(z_1, z_2)$ . Let  $z_i = (h_i, d_i)$  with  $a_i = |d_i| > 0$ . The inner minimization admits the explicit map evaluations given in Eq. (6), the gap between the predictions strictly vanishes

$$T_{\lambda, \mathcal{W}_2^2}^+(z_1) - T_{\lambda, \mathcal{W}_2^2}^-(z_2) = \frac{h_1 - h_2}{1 + \lambda(a_1 + a_2)} \xrightarrow{\lambda \rightarrow \infty} 0, \quad (13)$$

and moreover, both maps naturally converge to the weighted barycenter

$$T_{\lambda, \mathcal{W}_2^2}^+(z_1), T_{\lambda, \mathcal{W}_2^2}^-(z_2) \xrightarrow{\lambda \rightarrow \infty} \frac{a_2 h_1 + a_1 h_2}{a_1 + a_2} =: y^*(z_1, z_2), \quad (14)$$

where  $y^*(z_1, z_2)$  is the minimizer of the unrelaxed two-to-one cost  $\inf_y \{c_u(z_1, y) + c_u(z_2, y)\}$ .

At the level of pairwise costs, the closed form expression of the cost yields the pointwise convergence

$$C_{\lambda, c_2}^u(z_1, z_2) = \frac{\lambda}{1 + \lambda(a_1 + a_2)} (h_1 - h_2)^2 \xrightarrow{\lambda \rightarrow \infty} \frac{(h_1 - h_2)^2}{a_1 + a_2} =: C_{\infty}(z_1, z_2),$$

which is the unrelaxed OT cost.

Let now  $\pi_{\lambda}^* \in \Pi(\mu_{\eta, \Delta}^+, \mu_{\eta, \Delta}^-)$  be an optimal coupling for the reduced problem in Eq. (11). Denote by  $V_{\lambda}$  the optimal value in Eq. (11). Then  $V_{\lambda}$  is nondecreasing in  $\lambda$  and satisfies  $V_{\lambda} \leq V_{\infty}$ , where  $V_{\infty}$  is the optimal value of the unrelaxed problem (since  $C_{\lambda, c_2}^u \leq C_{\infty}$ ). We claim that we actually have

$$V_{\lambda} \xrightarrow{\lambda \rightarrow \infty} V_{\infty}.$$

Indeed, if were not the case, we could find a subsequence  $(\lambda_k)$  such that  $V_{\lambda}$  converges to some smaller limit along this subsequence. As the set  $\Pi(\mu_{\eta, \Delta}^+, \mu_{\eta, \Delta}^-)$  is compact for the weak convergence [32, Lemma 4.4], we may find a transport plan  $\pi_{\infty}^* \in \Pi(\mu_{\eta, \Delta}^+, \mu_{\eta, \Delta}^-)$  such that a subsequence of  $\pi_{\lambda_k}^*$  converges weakly towards  $\pi_{\infty}^*$ . Let us still denote this subsequence by  $\pi_{\lambda_k}^*$  for the sake of simplicity. We can then apply Fatou's lemma [15] to find that

$$V_{\infty} > \liminf_{k \rightarrow \infty} V_{\lambda_k} = \liminf_{k \rightarrow \infty} \int C_{\lambda_k, c_2}^u d\pi_{\lambda_k}^* \geq \int C_{\infty} d\pi_{\infty}^* \geq V_{\infty},$$

which yields a contradiction.

Finally, define the associated optimal outputs by

$$Y_{1, \lambda} := T_{\lambda, \mathcal{W}_2^2}^+(Z_1), \quad Y_{2, \lambda} := T_{\lambda, \mathcal{W}_2^2}^-(Z_2), \quad (Z_1, Z_2) \sim \pi_{\lambda}^*,$$

so that  $\nu_{1, \lambda}^* = \mathcal{L}(Y_{1, \lambda}) = (T_{\lambda, \mathcal{W}_2^2}^+)_\# \mu_{\eta, \Delta}^+$  and  $\nu_{2, \lambda}^* = \mathcal{L}(Y_{2, \lambda}) = (T_{\lambda, \mathcal{W}_2^2}^-)_\# \mu_{\eta, \Delta}^-$ . Consider moreover the coupling between these two output laws induced by the same construction,

$$\gamma_{\lambda} := (T_{\lambda, \mathcal{W}_2^2}^+, T_{\lambda, \mathcal{W}_2^2}^-)_\# \pi_{\lambda}^* \in \Pi(\nu_{1, \lambda}^*, \nu_{2, \lambda}^*).$$

By definition of  $\mathcal{W}_2$ , we have

$$\begin{aligned}\mathcal{W}_2^2(\nu_{1,\lambda}^*, \nu_{2,\lambda}^*) &\leq \int_{\mathbb{R}^2} (y_1 - y_2)^2 d\gamma_\lambda(y_1, y_2) = \mathbb{E}_{\pi_\lambda^*}[(Y_{1,\lambda} - Y_{2,\lambda})^2] \\ &= \int \frac{(h_1 - h_2)^2}{(1 + \lambda(|d_1| + |d_2|))^2} d\pi_\lambda^*(z_1, z_2) \\ &\leq \frac{2}{\lambda} \left( \int \frac{h_1^2}{|d_1|} d\mu_{\eta,\Delta}^+(z_1) + \int \frac{h_2^2}{|d_2|} d\mu_{\eta,\Delta}^-(z_2) \right),\end{aligned}$$

where we use the inequality  $(h_1 - h_2)^2 \leq 2h_1^2 + 2h_2^2$ . Since  $\int \frac{h^2}{|d|} d\mu_{\eta,\Delta}^\pm(z) = \int_{\mathcal{X}_\pm} \eta(x)^2 d\chi(x) \leq \mathbb{E}[\eta(X)^2] < \infty$ , we have  $\mathcal{W}_2^2(\nu_{1,\lambda}^*, \nu_{2,\lambda}^*) \rightarrow 0$  as  $\lambda \rightarrow \infty$ . In particular, the two output laws become arbitrarily close in  $\mathcal{W}_2$ .

Therefore the relaxed solutions asymptotically enforce a single common output distribution; if the unrelaxed barycenter is unique, this common limit must coincide with the unrelaxed barycenter law.

#### A.4 Proof of Proposition 2 (OT-U TV)

**Proposition 2 (OT-U TV).** *Under Assumption 1, Problem  $(P_{\lambda, \text{TV}})$  is equivalent to the OT problem*

$$\pi^* \in \arg \min_{\pi \in \Pi(\mu_{\eta,\Delta}^+, \mu_{\eta,\Delta}^-)} \int_{\mathbb{R}^2 \times \mathbb{R}^2} C_{\lambda, c_0}^u(z_1, z_2) d\pi(z_1, z_2), \quad (7)$$

where  $C_{\lambda, c_0}^u(z_1, z_2) = \min\left(\lambda, \frac{(h_1 - h_2)^2}{|d_1| + |d_2|}\right)$ . Moreover, a solution to the fair regression problem in Eq. (3) is given by Eq. (4) with

$$\begin{cases} T_{\lambda, \text{TV}}^+(z_1) = T_{\lambda, \text{TV}}^-(z_2) = \frac{|d_2|h_1 + |d_1|h_2}{|d_1| + |d_2|} & \text{if } \frac{(h_1 - h_2)^2}{|d_1| + |d_2|} \leq \lambda, \\ T_{\lambda, \text{TV}}^+(z_1) = h_1, \quad T_{\lambda, \text{TV}}^-(z_2) = h_2 & \text{if } \frac{(h_1 - h_2)^2}{|d_1| + |d_2|} > \lambda, \end{cases} \quad (8)$$

for all pairs  $z_1 = (h_1, d_1)$  and  $z_2 = (h_2, d_2)$  in the support of  $\pi^*$ .

*Proof of Proposition 2.* By Lemma 2, the global unaware problem reduces to evaluating the pointwise relaxed cost  $C_{\lambda, c_0}^u(z_1, z_2)$ . For any fixed pair  $z_1 = (h_1, d_1)$  and  $z_2 = (h_2, d_2)$  with absolute density weights  $a_1 = |d_1| > 0$  and  $a_2 = |d_2| > 0$ , we seek to minimize the objective

$$\Phi(y_1, y_2) = \frac{(h_1 - y_1)^2}{a_1} + \frac{(h_2 - y_2)^2}{a_2} + \lambda \mathbb{1}_{\{y_1 \neq y_2\}}.$$

We divide the pointwise optimization into two mutually exclusive cases depending on the indicator function.

*Case 1: We restrict the domain to  $y_1 \neq y_2$ .*

The indicator evaluates to 1, yielding the objective  $\Phi_{y_1 \neq y_2}(y_1, y_2) = \frac{(h_1 - y_1)^2}{a_1} + \frac{(h_2 - y_2)^2}{a_2} + \lambda$ . Because  $a_1, a_2 > 0$ , the unconstrained minimum of this convex quadratic is trivially achieved at  $y_1^* = h_1$  and  $y_2^* = h_2$ . Assuming  $h_1 \neq h_2$ , this solution respects the domain restriction, and the minimum cost for this branch is exactly  $\lambda$ .

*Case 2: We restrict the domain to  $y_1 = y_2 = y$ .*

The indicator evaluates to 0, simplifying the objective to a single-variable quadratic  $\Phi_{y_1 = y_2}(y) = \frac{(h_1 - y)^2}{a_1} + \frac{(h_2 - y)^2}{a_2}$ . The first-order optimality condition is  $-\frac{2(h_1 - y)}{a_1} - \frac{2(h_2 - y)}{a_2} = 0$ . Solving for  $y$  yields the weighted barycenter

$$y^* = \frac{(1/a_1)h_1 + (1/a_2)h_2}{1/a_1 + 1/a_2} = \frac{a_2 h_1 + a_1 h_2}{a_1 + a_2}. \quad (15)$$

We now plug  $y^*$  back into the objective

$$\Phi(y^*, y^*) = \frac{1}{a_1} \left( h_1 - \frac{a_2 h_1 + a_1 h_2}{a_1 + a_2} \right)^2 + \frac{1}{a_2} \left( h_2 - \frac{a_2 h_1 + a_1 h_2}{a_1 + a_2} \right)^2.$$

Simplifying the terms inside the squares gives  $\frac{a_1(h_1-h_2)}{a_1+a_2}$  for the first and  $\frac{-a_2(h_1-h_2)}{a_1+a_2}$  for the second. Squaring these and multiplying by their respective outer weights yields

$$\begin{aligned} \Phi(y^*, y^*) &= \frac{1}{a_1} \frac{a_1^2(h_1-h_2)^2}{(a_1+a_2)^2} + \frac{1}{a_2} \frac{a_2^2(h_1-h_2)^2}{(a_1+a_2)^2} \\ &= \frac{a_1+a_2}{(a_1+a_2)^2} (h_1-h_2)^2 \\ &= \frac{(h_1-h_2)^2}{a_1+a_2}. \end{aligned}$$

**Conclusion.** According to Lemma 2, a solution  $(\nu^{+*}, \nu^{-*})$  of  $(P_{\lambda, \mathcal{D}})$  is given by the marginals distributions of  $(y_1^*(Z_1, Z_2), y_2^*(Z_1, Z_2))$  where  $(Z_1, Z_2) \sim \pi^*$  and  $\pi^*$  is as in (7). Precisely, the map  $(y_1^*, y_2^*)$  is defined by letting  $(y_1^*(z_1, z_2), y_2^*(z_1, z_2)) = (h_1, h_2)$  if  $C_{\lambda, c_0}^u(z_1, z_2) = \lambda$ , and  $(y_1^*(z_1, z_2), y_2^*(z_1, z_2)) = (y^*, y^*)$  with  $y^*$  defined in (15) otherwise.

Our next goal is to show that under Assumption 1, the random variable  $y_1^*(Z_1, Z_2)$  does not depend on  $Z_2$ . We use once again Kantorovich duality.

Recall that there exists a function  $\psi$ , called a Kantorovich potential, that satisfies the following properties. Let  $\phi : z_2 \mapsto \sup_{z_1 \in \mathcal{Z}} (\psi(z_1) - C_{\lambda, c_0}(z_1, z_2))$  be the  $C_{\lambda, c_0}$ -transform of  $\psi$ . Then,  $\pi^*$  is supported on the  $C_{\lambda, c_0}^u$ -subdifferential of a Kantorovich potential  $\psi$ , that is the set

$$\Gamma = \{(z_1, z_2) \in \mathcal{Z} \times \mathcal{Z} : \psi(z_1) - \phi(z_2) = C_{\lambda, c_0}(z_1, z_2)\}.$$

Furthermore, the function  $\psi$  is  $C_{\lambda, c_0}$ -concave, meaning that for all  $z_1$ ,  $\psi(z_1) = \inf_{z_2} (\phi(z_2) + C_{\lambda, c_0}(z_1, z_2))$ . Using the definition of  $C_{\lambda, c_0}$ , we find that

$$\psi(z_1) = \inf_{z_2} (\phi(z_2) + \min(C_{\infty}(z_1, z_2), \lambda)) = \min \left( \tilde{\psi}(z_1), \inf_{z_2} \phi(z_2) + \lambda \right)$$

where  $\tilde{\psi}(z_1) = \inf_{z_2} (\phi(z_2) + C_{\infty}(z_1, z_2))$  defines a  $C_{\infty}$ -concave function for  $C_{\infty}(z_1, z_2) = \frac{(h_1-h_2)^2}{a_1+a_2}$ . According to Lemma 3 in [12], under Assumption 1, the function  $\tilde{\psi}$  is  $\mu_{\eta, \Delta}^+$ -almost everywhere differentiable at  $z_1 = (h_1, d_1)$  in the direction  $h_1$ .

Let  $\mathcal{Z}_{<} = \{z_1 \in \mathcal{Z} : \psi(z_1) = \tilde{\psi}(z_1)\}$ . If  $(z_1, z_2) \in \Gamma$  with  $z_1 \in \mathcal{Z}_{<}$ , then

$$\begin{aligned} C_{\infty}(z_1, z_2) &\geq \min(\lambda, C_{\infty}(z_1, z_2)) = C_{\lambda, c_0}(z_1, z_2) = \psi(z_1) - \phi(z_2) \\ &= \tilde{\psi}(z_1) - \phi(z_2) \geq C_{\infty}(z_1, z_2). \end{aligned}$$

Thus,  $\psi(z_1) - \phi(z_2) = C_{\infty}(z_1, z_2)$ . In other words, the set  $\{(z_1, z_2) \in \Gamma : z_1 \in \mathcal{Z}_{<}\}$  is included in the  $C_{\infty}$ -subdifferential of  $\tilde{\psi}$ . It is proven in [12] (as explained in the previous section) that if  $z_1$  is a point where  $\tilde{\psi}$  is differentiable at  $z_1$  in the direction  $h_1$ , and if  $(z_1, z_2)$  is in the  $C_{\infty}$ -subdifferential of  $\tilde{\psi}$ , then the quantity

$$y_1^*(z_1, z_2) = \frac{a_2 h_1 + a_1 h_2}{a_1 + a_2}$$

does not depend on the choice of  $z_2$ . Thus, the map  $T_{\lambda, \text{TV}}^+(z_1) = \frac{a_2 h_1 + a_1 h_2}{a_1 + a_2}$  is well-defined for  $\mu_{\eta, \Delta}^+$ -almost all  $z_1 \in \mathcal{Z}_{<}$ .



If  $z_1 \in \mathcal{Z}_= = \mathcal{Z} \setminus \mathcal{Z}_<$ , then  $\tilde{\psi}(z_1) > \inf_{z_2} \phi(z_2) + \lambda$ . Thus, if  $(z_1, z_2) \in \Gamma$ , then

$$\begin{aligned} \lambda &\geq \min(\lambda, C_\infty(z_1, z_2)) = C_{\lambda, c_0}(z_1, z_2) = \psi(z_1) - \phi(z_2) \\ &= \inf_{z'_2} \phi(z'_2) + \lambda - \phi(z_2) \geq \lambda. \end{aligned}$$

Thus  $C_{\lambda, c_0}(z_1, z_2) = \lambda$  and thus  $y_1^*(z_1, z_2) = h_1$  clearly only depends on  $z_1$ . This shows that the map  $T_{\lambda, \text{TV}}^+$  is well-defined  $\mu_{\eta, \Delta}^+$ -almost everywhere.

Similarly, we show that the map  $T_{\lambda, \text{TV}}^-$  is well-defined  $\mu_{\eta, \Delta}^-$ -almost everywhere.

We then conclude using Lemma 1 to show that solutions of Equation (3) take the form described in the lemma.  $\square$

Let us note that Proposition 1 implies in particular the result of Theorem 1 in the case  $c_D = c_0$ .

## A.5 Properties of the fairness penalty

Let  $m = \text{TV}(\chi^+, \chi^-)/2$ . Because the Total Variation between two probability measures is bounded by 2, we have  $m \leq 1$ . We begin by proving the stronger result that  $m\mathcal{U}_D(f) \geq D(\mathcal{L}(f(X)|S=+), \mathcal{L}(f(X)|S=-))$ . Since  $m \leq 1$  this immediately implies the looser bound  $\mathcal{U}_D(f) \geq D(\mathcal{L}(f(X)|S=+), \mathcal{L}(f(X)|S=-))$  stated in the claim. Recall that

$$\mathcal{U}_D(f) = \text{OT}_{c_D}(f_\#(\chi^+ - \chi^-)_+, f_\#(\chi^+ - \chi^-)_-).$$

To ease notations, let us denote  $\chi_{JD}^+ = m(\chi^+ - \chi^-)_+$ ,  $\chi_{JD}^- = m(\chi^+ - \chi^-)_-$ , and introduce the positive measure  $\tilde{\chi}_{JD} = \chi^- - \chi_{JD}^- = \chi^+ - \chi_{JD}^+$ . Then, we have  $\chi^+ = \chi_{JD}^+ + \tilde{\chi}_{JD}$  and  $\chi^- = \chi_{JD}^- + \tilde{\chi}_{JD}$ .

Now, denote  $\nu_{JD}^+ = f_\#\chi_{JD}^+$ ,  $\nu_{JD}^- = f_\#\chi_{JD}^-$ , and  $\tilde{\nu}_{JD} = f_\#\tilde{\chi}_{JD}$ . Similarly, denote  $\nu^+ = f_\#\chi^+$  and  $\nu^- = f_\#\chi^-$ . Since we have  $\chi^+ = \chi_{JD}^+ + \tilde{\chi}_{JD}$  and  $\chi^- = \chi_{JD}^- + \tilde{\chi}_{JD}$ , it must hold that  $\nu^+ = \nu_{JD}^+ + \tilde{\nu}_{JD}$  and  $\nu^- = \nu_{JD}^- + \tilde{\nu}_{JD}$ .

Let  $\pi$  be the optimal coupling solution to the problem  $\text{OT}_{c_D}(f_\#(\chi^+ - \chi^-)_+, f_\#(\chi^+ - \chi^-)_-)$ . Then,  $m\pi$  is a coupling between  $\nu_{JD}^+$  and  $\nu_{JD}^-$ . Define  $\tilde{\pi} = (I_d, I_d)_\#\tilde{\nu}_{JD}$ , then

$$\begin{aligned} \int_{\mathbb{R} \times \mathbb{R}} c_D(y_1, y_2) d(m\pi + \tilde{\pi})(y_1, y_2) &= \int_{\mathbb{R} \times \mathbb{R}} c_D(y_1, y_2) dm\pi(y_1, y_2) + \int_{\mathbb{R} \times \mathbb{R}} c_D(y_1, y_2) d\tilde{\pi}(y_1, y_2) \\ &= m \int_{\mathbb{R} \times \mathbb{R}} c_D(y_1, y_2) d\pi(y_1, y_2) + 0 \\ &= m\mathcal{U}_D(f). \end{aligned}$$

On the other hand,  $m\pi + \tilde{\pi}$  is a coupling with marginals  $\nu^+, \nu^-$ , so

$$\int_{\mathbb{R} \times \mathbb{R}} c_D(y_1, y_2) d(m\pi + \tilde{\pi})(y_1, y_2) \geq \text{OT}_{c_D}(\nu^+, \nu^-) = D(\mathcal{L}(f(X)|S=+), \mathcal{L}(f(X)|S=-))$$

which proves our first claim.

Now, the second claim follows immediately by noticing that in the awareness setting,  $\text{TV}(\chi^+, \chi^-) = 2$  and  $\chi^+ = (\chi^+ - \chi^-)_+$ ,  $\chi^- = (\chi^+ - \chi^-)_-$ .

## B Proofs for the relaxed aware setting

As discussed in Section 2.1, the aware setting corresponds to the boundary case of Problem  $(P_{\lambda, \mathcal{D}})$ , where the sensitive attribute is observed, that is we observe  $(X, S)$  instead of  $X$ . In that case, the Jordan decomposition partitions the space into the demographic groups, yielding uniform absolute density weights  $a_1 = 1/p^+$  for group  $S = +$  and  $a_2 = 1/p^-$  for group  $S = -$ .

Then,  $\chi^+$  and  $\chi^-$  have separate supports, with  $\Delta(x, +) = 1/p^+$  for  $(x, +)$  in the support of  $\chi^+$ ,  $\Delta(x, -) = -1/p^-$  for  $(x, -)$  in the support of  $\chi^-$ . The measure  $\mu_{\eta, \Delta}^+$  is simply the law of  $(\eta(X, +), 1/p^+)$  where  $X \sim \chi^+$ , and  $\mu_{\eta, \Delta}^-$  is the law of  $(\eta(X, -), -1/p^-)$  where  $X \sim \chi^-$ .

The two-dimensional costs described in Proposition 1 and Proposition 2 now depend effectively only on  $(h_1, h_2)$ , that is these costs become one-dimensional. In that case, the solution of the corresponding optimal transport problem can be formulated in a simpler fashion. Let us describe it more precisely.

**Exact Aware Solution.** First, in the limit  $\lambda \rightarrow \infty$  (no penalization), as established by Chzhen et al. [9], the exact Wasserstein barycenter for the 1D aware setting admits a closed-form. The fair prediction  $f_\infty^*(x, s)$  for an individual is obtained by evaluating their relative rank within their own demographic group via the cumulative distribution function  $F_{\mu_\eta^s}$ , and mapping it to the weighted quantiles of both groups

$$f_\infty^*(x, s) = q^* \circ F_{\mu_\eta^s}(\eta(x, s)) := \left( p^+ Q_{\mu_\eta^+} + p^- Q_{\mu_\eta^-} \right) \circ F_{\mu_\eta^s}(\eta(x, s))$$

where  $q^* : t \in (0, 1) \mapsto p^+ Q_{\mu_\eta^+}(t) + p^- Q_{\mu_\eta^-}(t)$  is the barycenter map.

### B.1 Aware $\mathcal{W}_2$ relaxation

Here we consider the fairness penalty  $\mathcal{D}(\cdot, \cdot) = OT_{c_2}(\cdot, \cdot)$  for  $c_2(y, y') = (y - y')^2$ .

**Corollary 1 (OT-A  $\mathcal{W}_2$ ).** *Under Assumption 1, in the awareness setting  $S$ , the solution to (3) is given by the optimal predictor  $f_{\lambda, \mathcal{W}_2}^*(x, s)$ , which interpolates between the unconstrained predictor  $\eta(x, s)$  and the exact fair predictor  $f_\infty^*(x, s)$*

$$f_{\lambda, \mathcal{W}_2}^*(x, s) = (1 - \alpha) f_\infty^*(x, s) + \alpha \eta(x, s), \quad \text{where} \quad \alpha = \frac{p^+ p^-}{p^+ p^- + \lambda}.$$

*This explicitly recovers the closed-form geodesic interpolation derived by Chzhen and Schreuder [8].*

*Proof.* In the framework of Chzhen and Schreuder [8], the fairness penalty is defined as the weighted variance  $p^+ p^- \mathcal{W}_2^2(\nu^+, \nu^-)$ . Thus, our  $\lambda$ -penalized objective is equivalent to their formulation under the effective penalty  $\lambda_C = \lambda/(p^+ p^-)$ .

According to their Proposition 4.7, this problem admits a relative improvement governed by an interpolation parameter  $\rho = (1 + \lambda_C)^{-2}$ . Following their Proposition 4.1, the optimal relaxed predictor is uniquely defined as the convex combination of the perfectly fair projection and the unconstrained base predictor, weighted by  $1 - \sqrt{\rho}$  and  $\sqrt{\rho}$  respectively. Substituting our penalty yields exactly

$$\sqrt{\rho} = \left( 1 + \frac{\lambda}{p^+ p^-} \right)^{-1} = \frac{p^+ p^-}{p^+ p^- + \lambda} = \alpha.$$

We now show that our generalized unaware reduction perfectly recovers this geometric interpolation from the maps defined in Proposition 1. Substituting the deterministic aware weights  $a_1 = 1/p^+$  and  $a_2 = 1/p^-$ , the interpolation denominator becomes

$$1 + \lambda(a_1 + a_2) = 1 + \lambda \left( \frac{p^+ + p^-}{p^+ p^-} \right),$$

and because  $p^+ + p^- = 1$ , this simplifies to

$$1 + \lambda(a_1 + a_2) = 1 + \frac{\lambda}{p^+ p^-} = \alpha^{-1}.$$

Thus,  $C_{\lambda, c_2}^u(z_1, z_2) = \alpha\lambda(h_1 - h_2)^2$  for  $z_1 = (h_1, 1/p^+)$  and  $z_2 = (h_2, -1/p^-)$ .

The optimal transport problem minimizing this cost is the monotone matching, obtained by letting  $h_1 = Q_{\mu_\eta^+}(t)$  and  $h_2 = Q_{\mu_\eta^-}(t)$  for a given  $t \in (0, 1)$ .

We then find from Eq. (6)

$$\begin{aligned} T_{\lambda, \mathcal{W}_2^2}^+(z_1) &= \alpha \left[ \left( 1 + \frac{\lambda}{p^-} \right) Q_{\mu_\eta^+}(t) + \frac{\lambda}{p^+} Q_{\mu_\eta^-}(t) \right] \\ &= \alpha Q_{\mu_\eta^+}(t) + \alpha\lambda \left( \frac{Q_{\mu_\eta^+}(t)}{p^-} + \frac{Q_{\mu_\eta^-}(t)}{p^+} \right). \end{aligned}$$

Recognizing from the definition of  $\alpha$  that  $\alpha\lambda = (1 - \alpha)p^+p^-$ , the second term simplifies to  $(1 - \alpha)(p^+Q_{\mu_\eta^+}(t) + p^-Q_{\mu_\eta^-}(t))$ , which corresponds to  $(1 - \alpha)q^*(t)$ , where  $q^*$  is the strict Wasserstein barycenter.

Evaluating this optimal quantile mapping at the individual's relative rank  $t_{x,+} = F_{\mu_\eta^+}(\eta(x, +))$  recovers the predictor formulation for the positive group

$$f_{\lambda, \mathcal{W}_2}^*(x, +) = (1 - \alpha)f_\infty^*(x, +) + \alpha\eta(x, +),$$

and symmetrically for group  $S = -$  using  $T^-$ . Thus, the unaware mapping retrieves the aware geodesic interpolation.  $\square$

## B.2 Aware TV relaxation

Here we consider the fairness penalty  $\mathcal{D}(\cdot, \cdot) = OT_{c_0}(\cdot, \cdot)$  for  $c_0(y, y') = \mathbb{1}_{\{y \neq y'\}}$ .

**Corollary 2 (OT-A TV).** *Under Assumption 1, in the awareness setting  $S \in \{+, -\}$ , the solution to (3) is characterized by the optimal transport plan  $\pi^* \in \Pi(\mu_\eta^+, \mu_\eta^-)$  that minimizes the cost*

$$\pi^* \in \arg \min_{\pi \in \Pi(\mu_\eta^+, \mu_\eta^-)} \int_{\mathbb{R} \times \mathbb{R}} \min(\lambda, p^+p^-(h_1 - h_2)^2) d\pi(h_1, h_2). \quad (16)$$

For any pair  $(h_1, h_2)$  in the support of  $\pi^*$ , the solution to Eq. (3) is given by

$$\begin{cases} f_{\lambda, \text{TV}}^*(h_1, +) = f_{\lambda, \text{TV}}^*(h_2, -) = p^+h_1 + p^-h_2 & \text{if } p^+p^-(h_1 - h_2)^2 \leq \lambda, \\ f_{\lambda, \text{TV}}^*(h_1, +) = h_1, \quad f_{\lambda, \text{TV}}^*(h_2, -) = h_2 & \text{if } p^+p^-(h_1 - h_2)^2 > \lambda. \end{cases}$$

*Proof of Corollary 2.* In the aware setting, the absolute density weights are constant and strictly determined by the group priors:  $a_1 = |d_1| = \frac{1}{p^+}$  for the positive group and  $a_2 = |d_2| = \frac{1}{p^-}$  for the negative group. We derive the aware solution by substituting these values directly into the results of Proposition 2.

First, we evaluate the denominator present in both the cost and the barycenter formulations

$$a_1 + a_2 = \frac{1}{p^+} + \frac{1}{p^-} = \frac{1}{p^+p^-},$$

where we used that  $p^+ + p^- = 1$ . Substituting this into the threshold from Proposition 2 yields

$$\frac{(h_1 - h_2)^2}{a_1 + a_2} = p^+p^-(h_1 - h_2)^2.$$

This recovers the threshold  $p^+p^-(h_1 - h_2)^2 \leq \lambda$  and the truncated cost  $\min(\lambda, p^+p^-(h_1 - h_2)^2)$  stated in the corollary.

Next, we substitute the weights into the optimal weighted barycenter  $y^*$  defined in Proposition 2

$$y^* = \frac{a_2 h_1 + a_1 h_2}{a_1 + a_2} = \left( \frac{1}{p^-} h_1 + \frac{1}{p^+} h_2 \right) p^+ p^- = p^+ h_1 + p^- h_2.$$

Applying this simplified cost threshold and this simplified barycenter directly into the piecewise formulation of the deterministic maps from Proposition 2 exactly recovers the closed-form predictions for  $f_{\lambda, \text{TV}}^*$ , concluding the proof.

Finally, we reconstruct the optimal map. According to Proposition 2, when the cost exceeds the threshold  $\lambda$ , the map outputs the unconstrained base prediction  $h_s$ . When the cost is below the threshold  $\lambda$ , the maps output the shared barycenter, recovering the formula in the statement of the corollary.  $\square$

## C Algorithm pseudocode

A pseudocode of our algorithm is given in Algorithm 1.

---

### Algorithm 1 Unaware Fair Regression

---

- 1: **Input:** Base regressor  $\hat{\eta}(x)$ , sensitive attribute's classifier  $\hat{p}(s|x)$ , priors  $p^+$  and  $p^-$ , dataset  $\{x_k\}_{k=1}^n$
  - 2: **Output:** Fair continuous map  $\hat{f}_{fair}$
  - 3: **Step 1: Estimation and Jordan decomposition**
  - 4: For all  $k \in \{1, \dots, n\}$ , compute unconstrained predictions  $\hat{\eta}_k \leftarrow \hat{\eta}(x_k)$ .
  - 5: For all  $k \in \{1, \dots, n\}$ , compute density ratio differences:  $\hat{\Delta}_k \leftarrow \frac{\hat{p}(+|x_k)}{p^+} - \frac{\hat{p}(-|x_k)}{p^-}$
  - 6: Define index sets:  $\mathcal{I}^+ = \{i : \hat{\Delta}_i > \tau\}$  and  $\mathcal{I}^- = \{j : \hat{\Delta}_j < -\tau\}$ .
  - 7: Compute normalized probability weights  $a_i$  (for  $i \in \mathcal{I}^+$ ) and  $b_j$  (for  $j \in \mathcal{I}^-$ ) so  $\sum a_i = \sum b_j = 1$ .
  - 8: **Step 2: Relaxed Optimal Transport**
  - 9: Cost  $\mathbf{C}_{i,j} \leftarrow C_{\lambda, \mathcal{D}} \left( (\hat{\eta}_i, \hat{\Delta}_i), (\hat{\eta}_j, \hat{\Delta}_j) \right)$  (e.g., via Proposition 1 or Proposition 2)
  - 10: Pairwise targets  $m_{i,j}^+ \leftarrow T_{\lambda, \mathcal{D}}^+ \left( (\hat{\eta}_i, \hat{\Delta}_i), (\hat{\eta}_j, \hat{\Delta}_j) \right)$  and  $m_{i,j}^- \leftarrow T_{\lambda, \mathcal{D}}^- \left( (\hat{\eta}_i, \hat{\Delta}_i), (\hat{\eta}_j, \hat{\Delta}_j) \right)$
  - 11: Solve discrete OT plan  $\Pi$  between  $\hat{\mu}^+$  (weights  $a_i$ ) and  $\hat{\mu}^-$  (weights  $b_j$ ) using cost matrix  $\mathbf{C}$ .
  - 12: **Step 3: Construct the fair targets**
  - 13: Initialize pseudo-label vector  $f = (f_1, \dots, f_n)$ .
  - 14: **for**  $k = 1$  to  $n$  **do**
  - 15:   **if**  $k \in \mathcal{I}^+$  **then**
  - 16:      $f_k \leftarrow \frac{1}{a_k} \sum_{j \in \mathcal{I}^-} \Pi_{k,j} m_{k,j}^+$
  - 17:   **else if**  $k \in \mathcal{I}^-$  **then**
  - 18:      $f_k \leftarrow \frac{1}{b_k} \sum_{i \in \mathcal{I}^+} \Pi_{i,k} m_{i,k}^-$
  - 19:   **else**
  - 20:      $f_k \leftarrow \hat{\eta}_k$
  - 21:   **end if**
  - 22: **end for**
  - 23: **Step 4: Fit final regressor**
  - 24: Train the continuous map  $\hat{f}_{fair}$  on inputs  $\{(\hat{\eta}_k, \hat{\Delta}_k)\}$  to predict targets  $\{f_k\}$ .
-

## D Experimental setup and implementation details

To ensure a rigorous, reproducible, and equitable comparison between our Optimal Transport framework and the FairReg baseline, we strictly adhere to the evaluation protocol established by Taturyan et al. [31].

### D.1 Details on the FairReg Baseline

Because FairReg [31] serves as the primary state-of-the-art baseline for relaxed unaware fairness throughout our experiments, it is necessary to detail its algorithmic mechanics to contextualize its performance and computational profile against our Optimal Transport framework.

**Grid discretization and dual optimization.** Unlike continuous Optimal Transport, FairReg operates as an unlabeled post-processing algorithm that does not map inputs to a deterministic continuous prediction. Instead, it relies on discretization, defining a uniform grid  $\hat{\mathcal{Y}}_L$  over a bounded interval  $[-B, B]$  consisting of  $2L + 1$  discrete points. The algorithm learns a randomized prediction function over this support, parameterized by a softmax function where the hyperparameter  $\beta$  controls the degree of entropic regularization. To enforce DP, they formulate a dual stochastic convex program from this entropically regularized objective. This dual problem is then minimized using a stochastic first-order oracle, typically implemented via Accelerated Stochastic Approximation (AC-SA2) to strictly control the expected norm of the gradient mapping.

**Iterations ( $T$ ).** In their theoretical framework and codebase, the hyperparameter  $T$  explicitly defines the total number of stochastic gradient evaluations. Because the stochastic gradient of the dual objective is computed by sampling exactly one independent unlabeled feature vector  $X$  per step,  $T$  represents individual iterations rather than full passes over the dataset. To guarantee a fair, scale-invariant convergence analysis across datasets of varying sizes ( $N$ ), we parameterize the baseline’s optimization in terms of training epochs. Specifically, we set  $T = 60 \times N$  for all experiments. This ensures the FairReg algorithm performs exactly 60 effective passes (epochs) over the unlabeled data, providing a rigorous and uniform optimization budget across all tasks.

### D.2 General experimental details

**Data preprocessing and splits.** For all real-world datasets, the continuous target variable  $Y$  is scaled linearly to the interval  $[-1, 1]$ . For all experiments, we evaluate out-of-sample generalization by splitting the data into a training set and a hold-out test set (using an 80% – 20% split), previously stratified by the sensitive attribute  $S$  to preserve group prior probabilities.

**Base models and estimators.** To isolate the effect of the fairness interventions and ensure differences in performance are strictly attributable to the post-processing algorithms, all methods share the exact same base estimators. Specifically, the unconstrained base regressor  $\hat{\eta}(X)$  is instantiated as a standard Ordinary Least Squares (OLS) Linear Regression model (we provide additional base model ablations study in Appendix F.3). The probabilistic sensitive attribute classifier  $\hat{\mathbb{P}}(S | X)$ , which is required by both FairReg and our Unaware OT framework to estimate the local density ratios, is instantiated as a Logistic Regression model trained for up to 2000 iterations to ensure convergence.

**Hyperparameters.** For the FairReg baseline, we follow their original implementation, learning a randomized prediction rule over a grid. For each relaxation level, the hyperparameters  $L$  (grid resolution) and  $\beta$  (entropy temperature) are selected via their proposed automated heuristic. For our OT framework, the discrete optimal transport plan  $\pi^*$  is solved exactly. To construct the final continuous, out-of-sample predictor  $\hat{f}_{fair}$  from the discrete OT assignments, we train a

Random Forest regressor (with 200 trees) to map the 2-dimensional feature space  $(\hat{\eta}(x), \hat{\Delta}(x))$  to the optimal fair targets  $y^*$ . The parameter  $\tau$  in 1 is set to the value  $10^{-6}$ .

**Statistical significance and error bars.** To assess the stability and statistical significance of our results, all experiments are aggregated over multiple independent trials (10 runs unless otherwise specified). The source of random variability depends on the experimental setting: for the synthetic dataset (Appendix E.2), the random seed dictates the underlying data generation process; for real-world datasets (Appendix F and Section 4), the random seed controls the stratified train/validation/test splits. In all summary tables, the reported  $\pm$  values represent the empirical standard deviation calculated across these independent runs (Appendices F.2, F.3 and G). Similarly, in all visualizations (such as the accuracy-fairness trade-off and confounding sweep curves), the shaded regions represent  $\pm 1$  standard deviation.

## E Experiments on synthetic datasets

We use two distinct synthetic datasets to empirically illustrate the exact geometric behavior of the optimal transport relaxations in both the aware and unaware settings: a non-linear confounded dataset designed to visualize the effect of the relaxations on the individual predictions, and a linearly confounded dataset designed to sweep the confounding strength and evaluate the accuracy-fairness trade-off across different levels of bias.

### E.1 Empirical illustration of the geometric relaxations

**Experimental setup and data generation.** To isolate and visualize the geometric behavior of the optimal transport relaxations, we use a 1-dimensional synthetic dataset designed to introduce a controlled, non-linear confounding effect. We generate  $N = 5000$  independent samples according to the following data generating process:

- **Sensitive attribute:** We sample the binary group uniformly at random  $S \sim \text{Bernoulli}(0.5)$ , and define its centered version  $Z_S = 2(S - 0.5) \in \{-1, 1\}$ .
- **Confounded feature:** We construct the continuous feature  $X \in \mathbb{R}$  as a linear combination of the sensitive group and an independent Gaussian noise, maintaining unit marginal variance:  $X = \gamma Z_S + \sqrt{1 - \gamma^2} \mathcal{N}(0, 1)$ , where the confounding strength is set to  $\gamma = 0.6$ .
- **Target variable:** The continuous outcome  $Y$  follows a non-linear S-curve dependence on  $X$  with additive Gaussian noise:  $Y = 3.0 \tanh(1.5X) + \mathcal{N}(0, 0.5^2)$ .

To completely remove finite-sample estimation artifacts and evaluate the pure geometry of the optimal transport framework, we bypass empirical risk minimization for the base models. Instead, we use exact mathematical oracles for the continuous conditional expectation  $\eta(x) = 3.0 \tanh(1.5x)$  and the exact posterior probabilities  $\mathbb{P}(S = 1 \mid X = x) = \sigma\left(\frac{2\gamma x}{1 - \gamma^2}\right)$ , where  $\sigma(z) = \frac{1}{1 + e^{-z}}$  is the sigmoid function.

Finally, to ensure an equitable comparison between the  $\mathcal{W}_2$  and TV penalties across both the aware and unaware paradigms, we perform a numerical line search to identify the exact  $\lambda$  (or exact threshold  $\tau$  for the aware TV case) that explicitly matches a predefined fairness budget (e.g., preserving exactly 80%, 60%, or 20% of the initial unconstrained Wasserstein-2 unfairness).

**Qualitative analysis of the unaware relaxations** To complement the analysis of the marginal distributions provided in the main text, we visualize here the exact effect of the relaxations directly on the individual predictions.

As shown in Fig. 5, comparing the base ERM predictor against the optimal relaxed unaware predictors (matched to an 80% unfairness budget) highlights the fundamental geometric difference

between the two penalties. The  $\mathcal{W}_2$  penalty induces a global smooth interpolation, gently pushing all individuals closer to the fair center. Conversely, the TV penalty creates a flat plateau, resulting in an overlapping density spike in the center of the distribution while leaving the extreme predictions completely untouched.

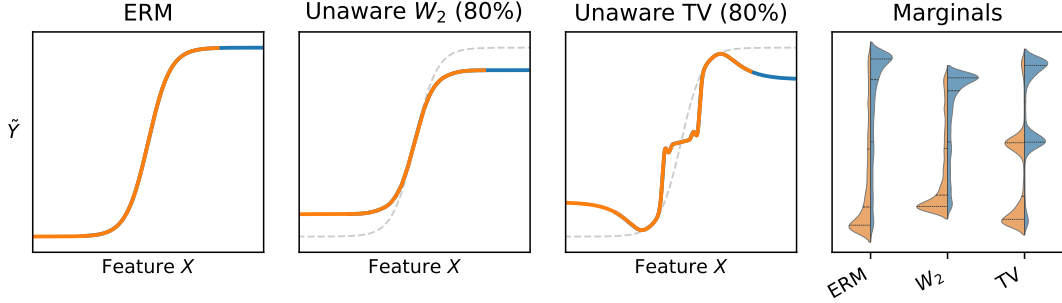


Figure 5: **Unaware relaxations.** Comparison of the Bayes predictor against the optimal relaxed unaware predictors under  $\mathcal{W}_2$  and TV penalties, matched to an 80% unfairness budget.

**Qualitative analysis of the aware relaxations** To complement the unaware visualizations above, we empirically evaluate the exact same relaxations in the *aware setting* on our 1D synthetic dataset.

As shown in Fig. 6, the aware  $\mathcal{W}_2$  penalty induces a smooth, proportional interpolation, shifting predictions homogeneously toward the central compromise. Conversely, the aware TV penalty acts as a strict hard-thresholding operator. This creates a highly discontinuous mapping: predictions perfectly merge if the pointwise disparity gap falls below the tolerated threshold, but strictly revert to the unconstrained biased baseline if the gap is too large.

By varying the regularization parameter to transition the model from the unconstrained empirical risk minimizer (100% unfair) to strict Demographic Parity (0% unfair), we observe the evolution of the marginal distributions in Fig. 7. Consequently, the aware TV penalty fragments the intermediate marginal distributions into distinct, multi-modal subgroups. This visually confirms its localized and discrete nature, standing in stark contrast to the continuous global compromise enforced by the  $\mathcal{W}_2$  penalty.

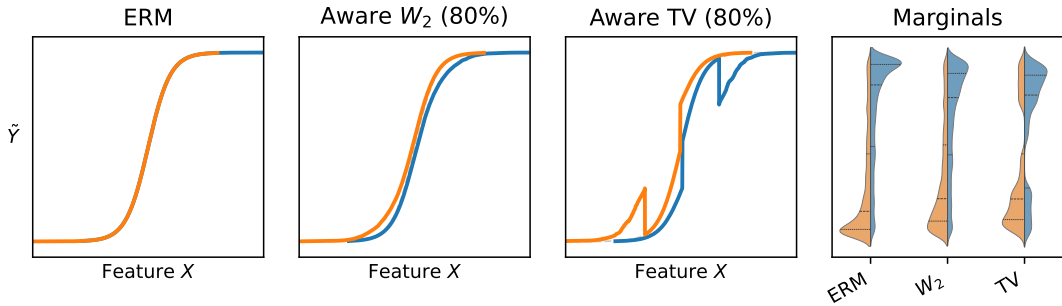


Figure 6: **Aware relaxations.** Comparison of the base ERM predictor against the optimal relaxed aware predictors under  $\mathcal{W}_2$  and TV penalties. While  $\mathcal{W}_2$  yields a smooth shift, the TV penalty applies an exact hard-thresholding, creating a discontinuous effect where predictions either perfectly merge or go back to the unfair ERM.

## E.2 Impact of confounding strenght and trade-offs

We generate  $n$  independent samples from a controlled 2-dimensional synthetic dataset.

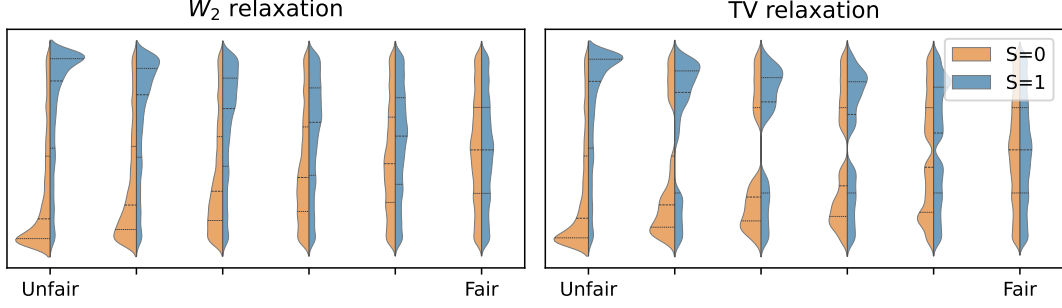


Figure 7: **Evolution of the aware marginal distributions.** Evolution of the predicted marginal distributions from Unfair to Fair. The discontinuous nature of the aware TV hard-thresholding creates fragmented, intermediate distributions, contrasting with the smooth continuous shift of the  $\mathcal{W}_2$  penalty.

**Sensitive attribute and features.** We sample the binary sensitive attribute uniformly at random  $S \sim \text{Bernoulli}(0.5)$ , and we define its centered version  $Z_S = 2(S - 0.5) \in \{-1, 1\}$ . We construct a 2-dimensional continuous feature vector  $X = (X_1, X_2)$ . The first feature,  $X_1$ , acts as the confounded signal and is generated exactly as described in Appendix E.1. The second feature,  $X_2$ , acts as a “safe” predictive signal and is drawn from an independent Gaussian distribution

$$X_2 \sim \mathcal{N}(0, 1).$$

**Target variable.** The continuous outcome  $Y$  is generated as a simple linear function of both features, with an independent additive Gaussian noise  $\varepsilon$

$$Y = 2X_1 - X_2 + \varepsilon, \quad \text{where } \varepsilon \sim \mathcal{N}(0, 0.5^2).$$

**Role of the parameter  $\gamma$ .** This specific design introduces a single source of unfairness governed by the parameter  $\gamma \in [0, 1]$ , which dictates the *confounding strength*. When  $\gamma = 0$ ,  $X_1$  is completely independent of  $S$ . As  $\gamma \rightarrow 1$ ,  $X_1$  becomes increasingly confounded with the demographic groups. Because  $Y$  is a function of  $X$ , the unconstrained ERM will always converge to the true coefficients  $[2.0, -1.0]$ , and its MSE will always equal the noise variance  $\text{Var}(\varepsilon) = 0.25$ , regardless of  $\gamma$ . This guarantees that any drop in accuracy observed during post-processing is exclusively attributable to the fairness constraints.

**Accuracy-fairness trade-offs.** We first fix  $\gamma = 0.5$  to induce a moderate structural bias and evaluate the relaxed predictors. As shown in Fig. 8, the unconstrained ERM achieves the theoretical optimal MSE of 0.25 but exhibits high unfairness. As we tighten the fairness constraints, OT Unaware recovers the Pareto frontier of the optimal aware interpolations. Notably, the naive plug-in estimator (OT  $\hat{S} \mathcal{W}_2$ ) fails to achieve the same fairness levels, demonstrating the severe risk of relying on a hard proxy classification  $\hat{S}$  rather than the continuous group probabilities.

**Impact of confounding strength.** To evaluate robustness across different biases, we fix the constraint to strict DP (no relaxation) and vary the confounding strength  $\gamma$  from  $10^{-2}$  to  $10^0$  (Figs. 9 and 10). Note that because the empirical  $\mathcal{W}_2$  distance is computed between finite samples, it exhibits a small, strictly positive baseline value due to sampling noise even for theoretically perfect fair models, which is why the models do not reach exactly zero on the logarithmic scale. This reveals a characteristic of the unaware geometry: it requires an active demographic signal to construct the 2D pseudo-measures. When  $\gamma$  is low,  $S$  is independent of  $X$ , causing the estimated



density ratio  $\hat{\Delta}(x)$  to vanish. The Unaware OT mapping conservatively collapses the predictions toward the global mean (incurring a high MSE) to strictly guarantee fairness.

However, as bias appears ( $\gamma > 0.1$ ), the unaware geometry fully resolves. In this high-confounding regime, OT Unaware suppresses the reliance on the biased feature while preserving the safe signal from  $X_2$ , converging perfectly to the optimal bounds of the Aware OT mapping. Furthermore, the naive plug-in approach (OT  $\hat{S} \mathcal{W}_2$ ) is not robust to this confounding, even at moderate levels. Because it forces a hard classification of the sensitive attribute, it confidently applies the optimal transport plan based on misaligned proxy labels. This results in a massive spike in true  $\mathcal{W}_2$  unfairness, only finally converging to the Aware bounds when  $\gamma \rightarrow 1$  and the proxy becomes a perfect oracle. Conversely, the FairReg baseline shows instability, with MSE penalties early in the trajectory before structural bias even materializes.

**Variance analysis.** This behavior at low confounding is explicitly confirmed by analyzing the within-group variance of the predictions (Fig. 11). At low  $\gamma$ , the variance of the OT Unaware predictions collapses significantly compared to the ERM, reflecting the prediction of the constant prediction in the absence of a reliable demographic signal. As the confounding signal strengthens ( $\gamma \rightarrow 1$ ), all fair models naturally reduce their variance to suppress  $X_1$ , but OT Unaware demonstrates the most stable and mathematically sound convergence toward the aware optimum.

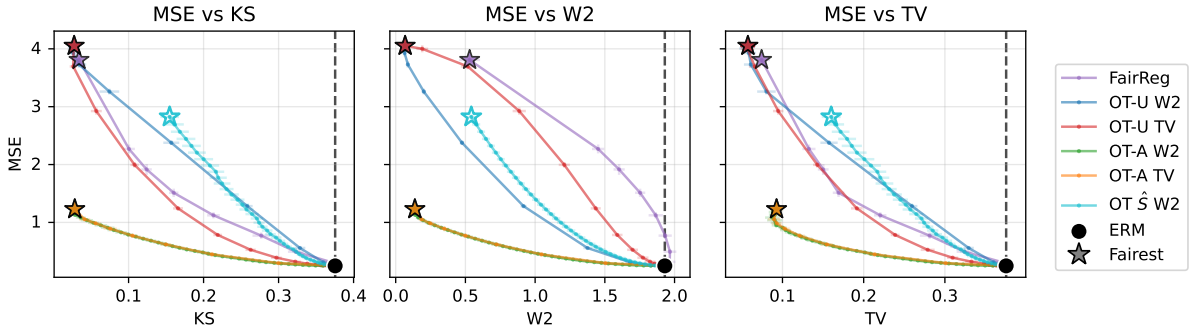


Figure 8: **Accuracy-fairness trade-offs on the synthetic dataset ( $\gamma = 0.5$ ).** Relaxation trajectories from the unconstrained empirical risk minimizer (ERM, black dot) to the strict fair models (stars). The unconstrained baseline naturally converges to the theoretical noise variance limit ( $\text{MSE} = 0.25$ ). Our exact optimal transport mappings (OT-U  $\mathcal{W}_2$  and OT-U TV) recover the optimal aware Pareto frontier and outperform both FairReg and the noisy plug-in proxy (OT  $\hat{S} \mathcal{W}_2$ ).

## F Experiments on real-world datasets

To complement the real-world evaluations in the main text, we provide the full accuracy-fairness trade-off analysis for the Adult (Census Income) dataset.

**Adult dataset detail.** The Adult dataset comprises individual-level records extracted from the 1994 U.S. Census database. It contains 48,842 total observations characterized by 14 demographic and employment-related attributes, including education level, marital status, occupation, and hours worked per week. While the usual prediction task for this dataset is binary classification (predicting whether an individual earns more than \$50K per year), [31] convert this into a continuous regression task by using a numeric proxy from the features (age) as the continuous target  $Y$ .

The dataset contains concrete demographic disparities: the population is approximately 66.9% male and 33.1% female. The racial composition of the dataset is also unbalanced, comprising

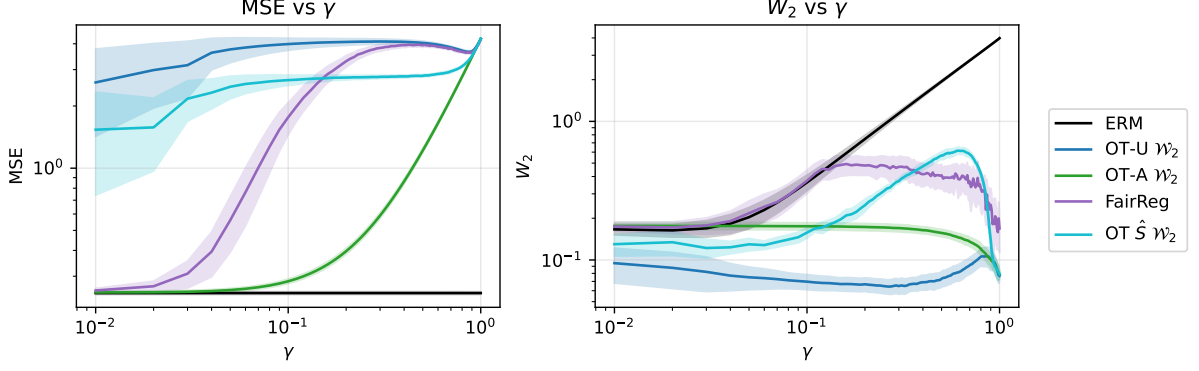


Figure 9: **Impact of confounding strength (Absolute)**. Evolution of the MSE (left) and  $\mathcal{W}_2$  unfairness (right) under strict DP. As  $\gamma \rightarrow 1$ , OT Unaware converges to the Aware model, tightly bounding the MSE degradation. FairReg suffers from optimization instability, triggering severe drops in predictive utility.

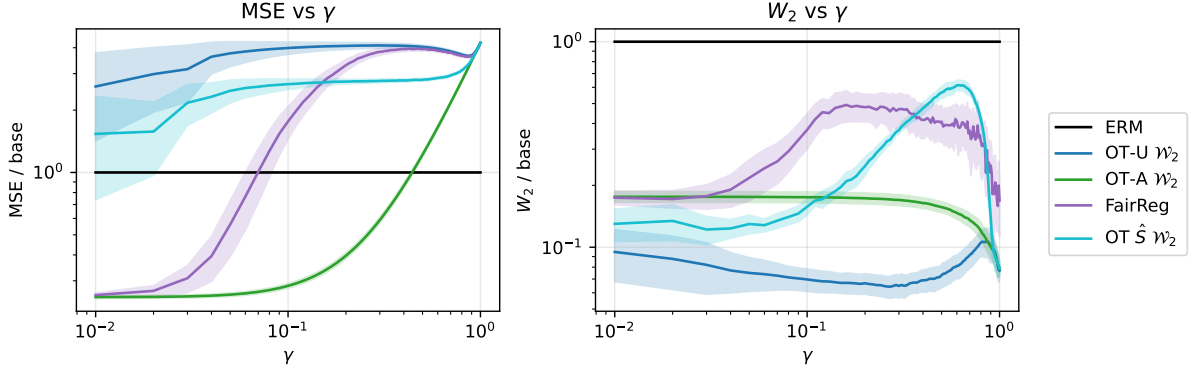


Figure 10: **Impact of confounding strength (Relative)**. Evolution of the MSE and  $\mathcal{W}_2$  unfairness expressed relative to the unconstrained ERM baseline. The relative visualization highlights FairReg’s rapid deterioration in utility at moderate confounding ( $\gamma \approx 0.1$ ), whereas OT Unaware maintains stability and accurately tracks the theoretical aware bound.

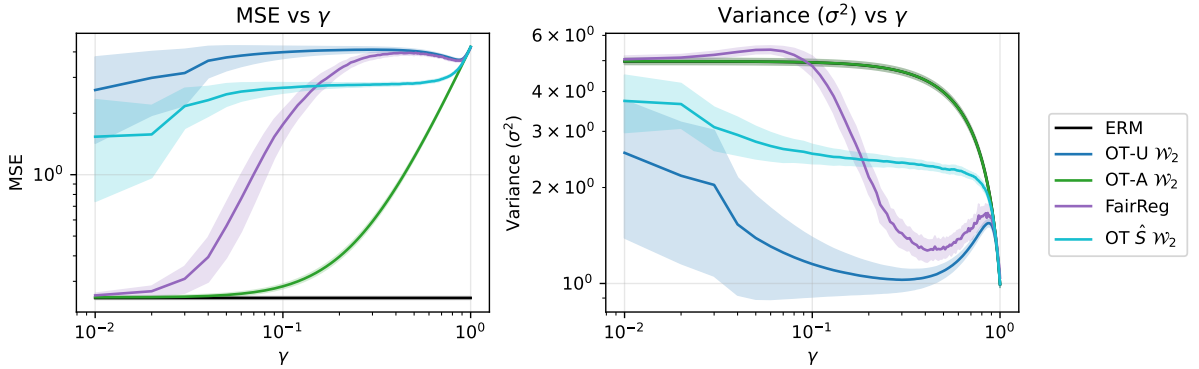


Figure 11: **Within-Group Variance Analysis**. Evolution of the MSE (left) and within-group prediction variance (right) across varying confounding strengths ( $\gamma$ ). At low  $\gamma$ , OT Unaware conservatively collapses the variance to act as a safe, constant predictor since the density ratio  $\hat{\Delta}(x)$  is uninformative. As structural bias increases, it successfully resolves the 2D geometry and recovers the variance profile of the optimal Aware mapping.

approximately 85.4% White, 9.6% Black, 3.2% Asian-Pac-Islander, 1.0% Amer-Indian-Eskimo, and 0.8% Other. For this experiment, the sensitive attribute considered is gender.

Characteristics about the **Law School** and **Communities and Crime** datasets are provided in the main text Section 4.

### F.1 Accuracy-fairness trade-offs

**Trade-off results.** As shown in Fig. 12, the OT Unaware framework consistently matches or outperforms the FairReg baseline across all fairness metrics ( $W_2$ , DP, and TV). The noisy plug-in estimator  $OT \hat{S} W_2$  fails at achieving fairness.

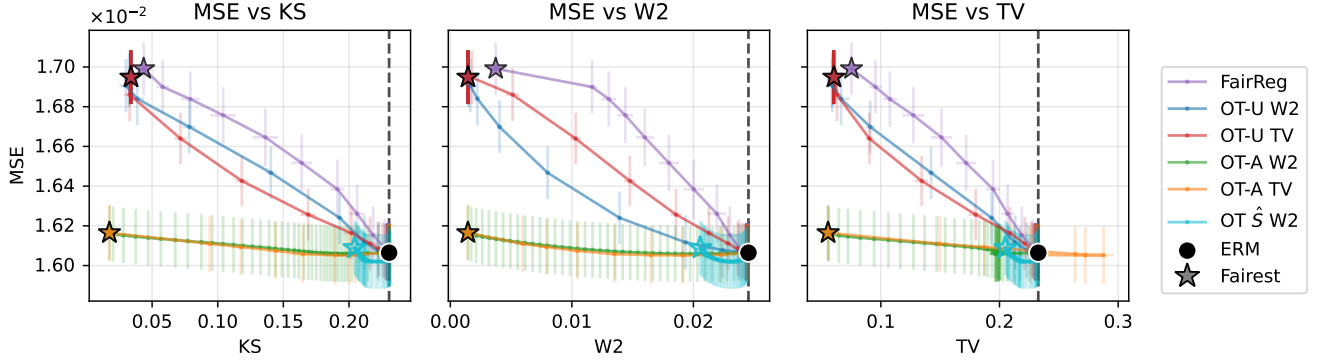


Figure 12: **Accuracy-fairness trade-offs on the Adult dataset.** Relaxation trajectories from the unconstrained empirical risk minimizer (ERM, black dot) to the exact fair models (stars). Error bars indicate the standard deviation across 10 random data splits. MSE of the constant predictor is  $1.69 \times 10^{-2}$ .

### F.2 Evaluation of the fully constrained models

**Experimental setup.** Tables 2 to 4 present the performance of the fully constrained models relative to the unconstrained ERM ( $MSE > 1.00$  indicates degradation; fairness metrics near 0.00 indicate perfect parity). Alongside FairReg and OT Unaware, we evaluate two naive plug-in baselines. If  $S$  is unavailable at inference, an intuitive solution is to substitute it with an estimator  $\hat{S}$  (or  $\hat{p}(s | x)$ ) inside the aware optimal transport maps  $T_s$ , yielding

$$\hat{Y}_{\text{hard}}(x) = T_{\hat{S}(x)}(\eta(x)), \quad (17)$$

$$\hat{Y}_{\text{soft}}(x) = \sum_s \hat{p}(s | x) T_s(\eta(x)). \quad (18)$$

**Results.** On the Adult dataset (Table 2), OT Unaware consistently matches or outperforms FairReg across all metrics. While FairReg achieves the lowest fairness measure in terms of KS-distance between the predictions of the group, at equality with our OT Unaware method, it still suffers from a high  $W_2^2$ -unfairness (23% relative to the baseline), whereas OT Unaware reduces it to 7% for the exact same relative MSE cost (1.06). OT Unaware also matches FairReg on TV and KS metrics, proving that exact optimal transport naturally guarantees statistical parity without the discretization.

**Failure of naive plug-in estimators.** The naive plug-in approaches completely fail to enforce fairness. On the Adult dataset (Table 2), the  $W_2$  and TV metrics for  $\hat{Y}_{\text{hard}}$  and  $\hat{Y}_{\text{soft}}$  remain between 0.85 and 1.10, making them virtually indistinguishable from the biased ERM. Because aware optimal transport relies on exact group rankings, classification errors from  $\hat{S}$  prevent

achieving fairness. In contrast, OT Unaware successfully enforces parity by mathematically absorbing this classification uncertainty directly into its global density weights, succeeding where plug-in estimators fails.

| Method            | MSE ↓                             | $W_2$ ↓                           | TV ↓                              | KS ↓                              | $KS^T$ ↓                          |
|-------------------|-----------------------------------|-----------------------------------|-----------------------------------|-----------------------------------|-----------------------------------|
| ERM Unfair        | $1.00 \pm 0.02$                   | $1.00 \pm 0.04$                   | $1.00 \pm 0.04$                   | $1.00 \pm 0.03$                   | $1.00 \pm 0.04$                   |
| FairReg           | $1.06 \pm 0.02$                   | $0.23 \pm 0.10$                   | $0.26 \pm 0.05$                   | <b><math>0.17 \pm 0.04</math></b> | <b><math>0.16 \pm 0.05</math></b> |
| OT Unaware        | $1.06 \pm 0.01$                   | <b><math>0.07 \pm 0.02</math></b> | <b><math>0.23 \pm 0.04</math></b> | <b><math>0.17 \pm 0.04</math></b> | <b><math>0.15 \pm 0.05</math></b> |
| OT $\hat{S}$ hard | <b><math>1.00 \pm 0.02</math></b> | $0.85 \pm 0.03$                   | $0.88 \pm 0.03$                   | $0.88 \pm 0.03$                   | $0.88 \pm 0.03$                   |
| OT $\hat{S}$ soft | $1.00 \pm 0.02$                   | $1.10 \pm 0.04$                   | $0.90 \pm 0.04$                   | $0.88 \pm 0.04$                   | $0.88 \pm 0.04$                   |
| OT aware          | $1.01 \pm 0.02$                   | $0.06 \pm 0.03$                   | $0.13 \pm 0.03$                   | $0.10 \pm 0.02$                   | $0.08 \pm 0.03$                   |

Table 2: **Relative performance on the Adult dataset.** Performance of the fully constrained models evaluated relative to the unconstrained ERM baseline. Values are reported as mean  $\pm$  standard deviation across 10 independent random splits. For fairness metrics ( $W_2$ , TV, KS,  $KS^T$ ), lower values indicate better parity; for MSE, values  $> 1.00$  indicate accuracy degradation. The best-performing unaware method is highlighted in **bold**, while statistically indistinguishable unaware methods ( $p > 0.05$  via Wilcoxon signed-rank test) are in **blue**. OT Unaware drastically improves geometric alignment ( $W_2$ ) over FairReg without incurring additional MSE degradation.

| Method            | MSE ↓                             | $W_2$ ↓                           | TV ↓                              | KS ↓                              | $KS^T$ ↓                          |
|-------------------|-----------------------------------|-----------------------------------|-----------------------------------|-----------------------------------|-----------------------------------|
| ERM Unfair        | $1.00 \pm 0.00$                   | $1.00 \pm 0.00$                   | $1.00 \pm 0.00$                   | $1.00 \pm 0.00$                   | $1.00 \pm 0.00$                   |
| FairReg           | $1.06 \pm 0.01$                   | $0.39 \pm 0.17$                   | <b><math>0.20 \pm 0.08</math></b> | <b><math>0.13 \pm 0.07</math></b> | <b><math>0.12 \pm 0.07</math></b> |
| OT Unaware        | $1.05 \pm 0.01$                   | <b><math>0.11 \pm 0.03</math></b> | <b><math>0.18 \pm 0.07</math></b> | <b><math>0.14 \pm 0.07</math></b> | <b><math>0.11 \pm 0.07</math></b> |
| OT $\hat{S}$ hard | $1.03 \pm 0.01$                   | $0.37 \pm 0.04$                   | $0.43 \pm 0.06$                   | $0.44 \pm 0.05$                   | $0.42 \pm 0.07$                   |
| OT $\hat{S}$ soft | <b><math>1.02 \pm 0.00</math></b> | $0.46 \pm 0.03$                   | $0.54 \pm 0.06$                   | $0.56 \pm 0.04$                   | $0.54 \pm 0.06$                   |
| OT aware          | $1.03 \pm 0.00$                   | $0.11 \pm 0.02$                   | $0.18 \pm 0.05$                   | $0.13 \pm 0.03$                   | $0.09 \pm 0.04$                   |

Table 3: **Relative performance on the Law School dataset.** Metrics are normalized relative to the unconstrained ERM baseline, averaged across 10 random splits. Formatting follows the same conventions as the Adult dataset. OT Unaware systematically achieves superior geometric alignment ( $W_2$ ) while matching the statistical parity (KS) of FairReg. In contrast, the naive plug-in estimators (OT  $\hat{S}$ ) completely fail to enforce meaningful fairness, demonstrating the fragility of relying on hard proxy labels.

### F.3 Robustness across different base estimators

To ensure the performance of the post-processing methods is not an artifact of the initial model class, we benchmark the fully constrained frameworks across three distinct unconstrained base estimators: Ordinary Least Squares (Linear), Random Forests (RF), and Gradient Boosting Machines (GBM).

**Methodology.** For each base estimator, we extract the unconstrained predictions (ERM) and process them using both the FairReg baseline and our OT Unaware framework. To ensure a rigorous comparison, we report the mean and standard deviation across 10 random train/test splits of size. For the two unaware methods, the strictly superior performance is highlighted in **bold**. To account for variance, we apply a Wilcoxon signed-rank test; if a method underperforms

| Method            | MSE ↓                             | $W_2$ ↓                           | TV ↓                              | KS ↓                              | $KS^{\mathcal{T}}$ ↓              |
|-------------------|-----------------------------------|-----------------------------------|-----------------------------------|-----------------------------------|-----------------------------------|
| ERM Unfair        | $1.00 \pm 0.00$                   | $1.00 \pm 0.00$                   | $1.00 \pm 0.00$                   | $1.00 \pm 0.00$                   | $1.00 \pm 0.00$                   |
| FairReg           | $1.99 \pm 0.21$                   | $0.20 \pm 0.08$                   | $0.37 \pm 0.06$                   | <b><math>0.16 \pm 0.04</math></b> | <b><math>0.16 \pm 0.03</math></b> |
| OT Unaware        | $1.92 \pm 0.20$                   | <b><math>0.09 \pm 0.04</math></b> | <b><math>0.35 \pm 0.08</math></b> | $0.18 \pm 0.08$                   | $0.16 \pm 0.08$                   |
| OT $\hat{S}$ hard | $1.73 \pm 0.20$                   | $0.22 \pm 0.09$                   | $0.43 \pm 0.06$                   | $0.27 \pm 0.07$                   | $0.25 \pm 0.08$                   |
| OT $\hat{S}$ soft | <b><math>1.50 \pm 0.17</math></b> | $0.27 \pm 0.09$                   | $0.42 \pm 0.05$                   | $0.36 \pm 0.07$                   | $0.34 \pm 0.07$                   |
| OT aware          | $1.66 \pm 0.19$                   | $0.13 \pm 0.04$                   | $0.38 \pm 0.04$                   | $0.17 \pm 0.03$                   | $0.14 \pm 0.04$                   |

Table 4: **Relative performance on the Communities and Crime dataset.** Performance normalized relative to the ERM baseline across 10 random splits. OT Unaware systematically dominates the FairReg baseline, achieving both a lower relative MSE (1.92 vs 1.99) and significantly superior geometric alignment ( $W_2=0.09$  vs 0.20). This demonstrates that FairReg struggles to enforce parity on highly confounded datasets without severely degrading predictive utility.

the best method but the difference is not statistically significant ( $p > 0.05$ ), it is highlighted in blue.

**Metrics.** We recall that  $KS^{\mathcal{T}}$  is a discretized approximation of the Kolmogorov-Smirnov (KS) distance, computed as the maximum absolute c.d.f. difference across a discrete 50-bin grid  $\mathcal{T}$ . We also report the continuous Wasserstein-2 ( $W_2$ ) and Total Variation (TV) distances. The Total Variation is computed over 50 bins.

**Base model configurations.** To ensure reproducibility, we use the following hyperparameters for our unconstrained base estimators (implemented via scikit-learn):

- **Linear:** Ordinary Least Squares with an intercept (and Logistic Regression with `max_iter=2000` for auxiliary classification).
- **Random Forest (RF):** 100 trees restricted to a minimum of 20 samples per leaf (`min_samples_leaf=20`) to prevent severe overfitting.
- **Gradient Boosting Machine (GBM):** 100 boosting stages with a maximum tree depth of 3 (`max_depth=3`).

**Performance and interpretation.** As shown in Tables 5 to 7, OT Unaware demonstrates robust generalization across all model classes. On the individual-level datasets (Adult and Law School), OT Unaware systematically achieves superior  $W_2$  and  $KS^{\mathcal{T}}$ .

On the Communities dataset (Table 7), OT Unaware proves even more dominant. Not only does it achieve a lower MSE than the FairReg baseline across all model classes (e.g., 0.043 vs 0.044 for Linear), but it also strictly enforces the geometric fairness constraint. FairReg struggles to align the distributions, leaving a  $W_2$  gap of 0.072 for the Linear model, whereas OT Unaware successfully reduces it to 0.028.

**Theoretical validation of the MSE degradation.** The main observation in the Adult and Law School results is that the empirical Mean Squared Error (MSE) appears virtually unchanged between the Unfair ERM and the strictly fair OT predictors (e.g., remaining at 0.01 or 0.02). This is not an anomaly, but a direct empirical validation of the theoretical bounds of Wasserstein fair regression.

The theoretical maximum MSE degradation required to achieve exact DP is strictly bounded by the squared Wasserstein distance ( $W_2^2$ ) between the initial biased distributions. For the Law School dataset, the initial  $W_2$  gap is approximately 0.03. The exact geometric cost to perfectly repair this disparity is therefore bounded by  $0.03^2 = 0.0009$ . Because the targets are scaled to  $[-1, 1]$ , the base ERM risk is naturally around 0.010. Adding the absolute maximum fairness penalty yields an expected fair risk of 0.0109. Consequently, the degradation is practically invisible at two decimal places. This proves that continuous Optimal Transport successfully enforces parity with the absolute minimum possible disruption to accuracy.

Conversely, for the Communities and Crime dataset, the initial baseline disparities are severe ( $W_2 \approx 0.31$ ). The theoretical geometric cost to achieve parity is therefore bounded by  $0.31^2 \approx 0.096$ . As expected, the empirical MSE degradation on this dataset is highly visible (jumping from 0.022 for the ERM to 0.043 for OT), perfectly tracking the quadratic cost bound.

| Base Model    | Method     | MSE ↓                | $W_2$ ↓              | $KS^T$ ↓             |
|---------------|------------|----------------------|----------------------|----------------------|
| <b>LINEAR</b> | ERM Unfair | 0.016 ± 0.000        | 0.024 ± 0.001        | 0.226 ± 0.010        |
|               | FairReg    | 0.017 ± 0.000        | 0.006 ± 0.002        | 0.037 ± 0.009        |
|               | OT Unaware | <b>0.017 ± 0.000</b> | <b>0.002 ± 0.000</b> | <b>0.035 ± 0.011</b> |
|               | OT aware   | 0.016 ± 0.000        | 0.002 ± 0.001        | 0.018 ± 0.006        |
| <b>RF</b>     | ERM Unfair | 0.015 ± 0.000        | 0.020 ± 0.001        | 0.166 ± 0.010        |
|               | FairReg    | 0.016 ± 0.000        | 0.004 ± 0.001        | 0.020 ± 0.005        |
|               | OT Unaware | <b>0.016 ± 0.000</b> | <b>0.003 ± 0.001</b> | <b>0.019 ± 0.006</b> |
|               | OT aware   | 0.015 ± 0.000        | 0.003 ± 0.001        | 0.019 ± 0.006        |
| <b>GBM</b>    | ERM Unfair | 0.015 ± 0.000        | 0.021 ± 0.001        | 0.199 ± 0.007        |
|               | FairReg    | 0.016 ± 0.000        | 0.006 ± 0.001        | 0.023 ± 0.006        |
|               | OT Unaware | <b>0.016 ± 0.000</b> | <b>0.004 ± 0.001</b> | 0.023 ± 0.009        |
|               | OT aware   | 0.015 ± 0.000        | 0.004 ± 0.001        | 0.020 ± 0.006        |

Table 5: **Robustness benchmark on the Adult dataset.** OT Unaware strictly dominates the baseline across all base models, achieving near-perfect parity ( $W_2$  drops to 0.002 for Linear) with negligible MSE degradation.

## G Computational efficiency

We report here the average training time (in seconds) for each method across 10 independent runs on a standard CPU (Intel Core i7-9700K @ 3.60GHz). As shown in Table 8, the OT Unaware framework achieves massive computational speedups compared to the FairReg baseline, which relies on iterative stochastic gradient optimization. For instance, on the Communities dataset, FairReg takes on average 9.97 seconds to train, while OT Unaware takes only 0.37 seconds, yielding a speedup of over  $26\times$ .

| Base Model | Method     | MSE ↓                | $W_2$ ↓              | $KS^T$ ↓             |
|------------|------------|----------------------|----------------------|----------------------|
| LINEAR     | ERM Unfair | 0.009 ± 0.000        | 0.028 ± 0.001        | 0.284 ± 0.015        |
|            | FairReg    | 0.010 ± 0.000        | 0.007 ± 0.002        | 0.036 ± 0.020        |
|            | OT Unaware | <b>0.010 ± 0.000</b> | <b>0.003 ± 0.001</b> | <b>0.032 ± 0.022</b> |
|            | OT aware   | 0.009 ± 0.000        | 0.003 ± 0.001        | 0.025 ± 0.011        |
| RF         | ERM Unfair | 0.008 ± 0.000        | 0.026 ± 0.002        | 0.210 ± 0.022        |
|            | FairReg    | 0.009 ± 0.000        | 0.011 ± 0.010        | 0.048 ± 0.013        |
|            | OT Unaware | <b>0.009 ± 0.000</b> | <b>0.004 ± 0.001</b> | <b>0.039 ± 0.011</b> |
|            | OT aware   | 0.008 ± 0.000        | 0.003 ± 0.001        | 0.026 ± 0.008        |
| GBM        | ERM Unfair | 0.008 ± 0.000        | 0.026 ± 0.002        | 0.190 ± 0.021        |
|            | FairReg    | 0.008 ± 0.000        | 0.010 ± 0.007        | 0.030 ± 0.011        |
|            | OT Unaware | <b>0.008 ± 0.000</b> | <b>0.003 ± 0.001</b> | <b>0.028 ± 0.014</b> |
|            | OT aware   | 0.008 ± 0.000        | 0.003 ± 0.001        | 0.025 ± 0.005        |

Table 6: **Robustness benchmark on the Law School dataset.** Mean and standard deviation of fairness and accuracy metrics are reported across 10 runs for fully constrained models. For the unaware methods (FairReg and OT Unaware), the best performing method is in **bold**. Statistically insignificant differences (Wilcoxon signed-rank test,  $p > 0.05$ ) are highlighted in blue.

| Base Model | Method     | MSE ↓                | $W_2$ ↓              | $KS^T$ ↓             |
|------------|------------|----------------------|----------------------|----------------------|
| LINEAR     | ERM Unfair | 0.022 ± 0.003        | 0.310 ± 0.024        | 0.624 ± 0.062        |
|            | FairReg    | 0.044 ± 0.003        | 0.072 ± 0.034        | 0.112 ± 0.031        |
|            | OT Unaware | <b>0.043 ± 0.003</b> | <b>0.028 ± 0.012</b> | <b>0.096 ± 0.042</b> |
|            | OT aware   | 0.037 ± 0.002        | 0.041 ± 0.013        | 0.090 ± 0.026        |
| RF         | ERM Unfair | 0.020 ± 0.001        | 0.281 ± 0.020        | 0.643 ± 0.055        |
|            | FairReg    | 0.044 ± 0.002        | 0.046 ± 0.026        | 0.129 ± 0.050        |
|            | OT Unaware | <b>0.044 ± 0.002</b> | <b>0.021 ± 0.010</b> | <b>0.105 ± 0.053</b> |
|            | OT aware   | 0.036 ± 0.002        | 0.026 ± 0.011        | 0.105 ± 0.048        |
| GBM        | ERM Unfair | 0.021 ± 0.001        | 0.296 ± 0.023        | 0.629 ± 0.061        |
|            | FairReg    | 0.042 ± 0.002        | 0.040 ± 0.015        | 0.119 ± 0.035        |
|            | OT Unaware | <b>0.040 ± 0.002</b> | <b>0.033 ± 0.008</b> | <b>0.117 ± 0.036</b> |
|            | OT aware   | 0.037 ± 0.002        | 0.042 ± 0.016        | 0.114 ± 0.049        |

Table 7: **Robustness benchmark on the Communities and Crime dataset.** OT Unaware systematically dominates the FairReg baseline across all model classes, achieving both a lower MSE and significantly superior geometric alignment. While FairReg struggles to enforce parity (leaving a  $W_2$  gap of 0.072 for Linear), OT Unaware successfully enforces the geometric constraint ( $W_2 = 0.028$ ) while incurring less predictive degradation.

| Dataset           | FairReg            | OT Unaware       | OT $\hat{S}$      | OT Aware          |
|-------------------|--------------------|------------------|-------------------|-------------------|
| Synthetic (N=10k) | $56.87 \pm 7.30$   | $8.75 \pm 1.26$  | $0.005 \pm 0.004$ | $0.003 \pm 0.002$ |
| Adult             | $167.23 \pm 8.22$  | $94.21 \pm 5.35$ | $0.034 \pm 0.012$ | $0.008 \pm 0.002$ |
| Communities       | $9.97 \pm 0.20$    | $0.37 \pm 0.01$  | $0.014 \pm 0.001$ | $0.008 \pm 0.000$ |
| Lawschool         | $117.68 \pm 16.18$ | $31.03 \pm 0.58$ | $0.053 \pm 0.011$ | $0.006 \pm 0.004$ |

Table 8: **Computational efficiency benchmark.** Average model training time (in seconds  $\pm$  standard deviation) evaluated across 10 independent runs. By relying on exact, closed-form geometric reductions, the OT Unaware framework bypasses the iterative stochastic gradient optimization required by FairReg, yielding massive computational speedups (e.g., up to  $26\times$  faster on the Communities dataset).